



Top 72 CAT Progressions and Series Questions With Video Solutions

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Questions

Instructions

For the following questions answer them individually

Question 1

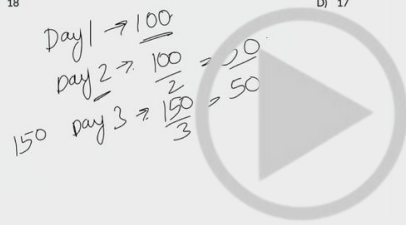
On day one, there are 100 particles in a laboratory experiment. On day n , where $n \geq 2$, one out of every n articles produces another particle. If the total number of particles in the laboratory experiment increases to 1000 on day m , then m equals

- A 19
- B 16
- C 18
- D 17

On day one, there are 100 particles in a laboratory experiment. On day n , where $n \geq 2$, one out of every n articles produces another particle. If the total number of particles in the laboratory experiment increases to 1000 on day m , then m equals

A) 19 B) 16
C) 18 D) 17

Day 1 $\rightarrow 100$
Day 2 $\rightarrow \frac{100}{2} = 50$
Day 3 $\rightarrow \frac{150}{3} = 50$



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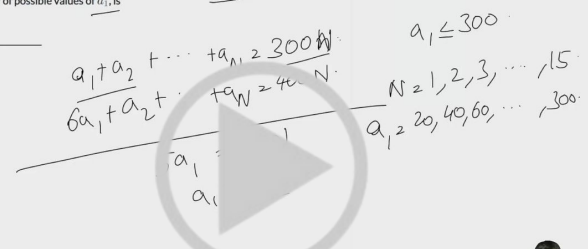
Question 2

The average of a non-decreasing sequence of N numbers $a_1, a_2, \dots, a_N$ is 300. If a_1 is replaced by $6a_1$, the new average becomes 400. Then, the number of possible values of a_1 is

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Answer: _____

$a_1 + a_2 + \dots + a_N = 300N$
 $6a_1 + a_2 + \dots + a_N = 400N$
 $\frac{a_1 + a_2 + \dots + a_N}{N} = 300$
 $\frac{6a_1 + a_2 + \dots + a_N}{N} = 400$
 $a_1 \leq 300$
 $N = 1, 2, 3, \dots, 15$
 $a_1 = 20, 40, 60, \dots, 300$



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Question 3

If the population of a town is p in the beginning of any year then it becomes $3 + 2p$ in the beginning of the next year. If the population in the beginning of 2019 is 1000, then the population in the beginning of 2034 will be

- A $(1003)^{15} + 6$
- B $(997)^{15} - 3$
- C $(997)2^{14} + 3$
- D $(1003)2^{15} - 3$

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If the population of a town is p in the beginning of any year then it becomes $3 + 2p$ in the beginning of the next year. If the population in the beginning of 2019 is 1000, then the population in the beginning of 2034 will be

Handwritten solution:

$$1 - p$$

$$2 - 2p + 3$$

$$3 - 4p + 3 \times 2 + 3 = 4p + 3 \times (1+2)$$

$$4 = 8p + 3 \times (2+4) + 3 = 10p + 2 \times (1+2+4)$$

$$5 = 16p + 3 \times (2+4+8) + 3 = 16p + 3 \times (2+4+8)$$

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Question 4

Consider the sequence of numbers $a_1, a_2, a_3, \dots$ to infinity where $a_1 = 81.33$ and $a_2 = -19$ and $a_j = a_{j-1} - a_{j-2}$ for $j \geq 3$. What is the sum of the first 6002 terms of this sequence?

- A -100.33
- B -30.00
- C 62.33
- D 119.33

cracku

Consider the sequence of numbers $a_1, a_2, a_3, \dots$ to infinity where $a_1 = 81.33$ and $a_2 = -19$ and $a_j = a_{j-1} - a_{j-2}$ for $j \geq 3$. What is the sum of the first 6002 terms of this sequence?

Handwritten solution:

$$a_3 = a_2 - a_1$$

$$a_4 = a_3 - a_2 = -a_1$$

$$a_5 = a_4 - a_3 = -a_1 - (-a_1) = 0$$

$$a_6 = a_5 - a_4 = 0 - (-a_1) = a_1$$

$$a_7 = a_6 - a_5 = a_1 - 0 = a_1$$

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Instructions

Answer the questions based on the following information. A series S_1 of five positive integers is such that the third term is half the first term and the fifth term is 20 more than the first term. In series S_2 , the n th term is defined as the difference between the $(n+1)$ th term and the n th term of series S_1 , S_2 is an arithmetic progression with a common difference of 30.

Question 5

First term of S_1 is

- A 80
- B 90
- C 100
- D 120

[VIEW SOLUTION](#)**Question 6**

What is the difference between fourth terms of S_1 and S_2 ?

- A 10
- B 20
- C 30
- D 40

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For the following questions answer them individually

Question 7

Let both the series $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ be in arithmetic progression such that the common differences of both the series are prime numbers. If $a_5 = b_9$, $a_{19} = b_{19}$ and $b_2 = 0$, then a_{11} equals

- A 86
- B 79
- C 83
- D 84

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Let both the series $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ be in arithmetic progression such that the common differences of both the series are prime numbers. If $a_5 = b_9$, $a_{19} = b_{19}$ and $b_2 = 0$, then a_{11} equals

A) 86
B) 79
C) 83
D) 84

$a_1 + 4p_1 = b_1 + 8p_2$
 $= 7p_2$
 $a_1 + 18p_1 = b_1 + 18p_2$
 $b_1 + p_2 = 0$

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Question 8

For a sequence of real numbers $x_1, x_2, \dots, x_n$, if $x_1 - x_2 + x_3 - \dots + (-1)^{n+1}x_n = n^2 + 2n$ for all natural numbers n , then the sum $x_{49} + x_{50}$ equals

- A 200
- B 2
- C -200
- D -2

cracku

For a sequence of real numbers $x_1, x_2, \dots, x_n$, if $x_1 - x_2 + x_3 - \dots + (-1)^{n+1}x_n = n^2 + 2n$ for all natural numbers n , then the sum $x_{49} + x_{50}$ equals

A) 200
B) 2
C) -200
D) -2

$x_1 - x_2 + x_3 - \dots + (-1)^{n+1}x_n = n^2 + 2n$
 $x_1 - x_2 + x_3 - \dots + (-1)^{49+1}x_{49} = 49^2 + 2 \times 49$
 $x_1 - x_2 + x_3 - \dots + (-1)^{50+1}x_{50} = 50^2 + 2 \times 50$
 $\underline{\hspace{1cm}}$
 $2x_{49} - x_{50} = 49^2 - 50^2 + 2 \times 49 - 2 \times 50$
 $= (49 - 50)(49 + 50) + 2 \times 49 - 2 \times 50$
 $= (-1)(99) + 98 - 100$
 $= -99 + 98 - 100$
 $= -101$

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Question 9

If $x_1 = -1$ and $x_m = x_{m+1} + (m + 1)$ for every positive integer m , then X_{100} equals

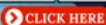
- A -5050
- B -5151
- C -5051
- D -5150

If $x_1 = -1$ and $x_m = x_{m+1} + (m+1)$ for every positive integer m , then x_{100} equals

A) -5050
B) -5151
C) -5051
D) -5150

$x_1 = -1$
 $x_{m+1} = x_m - (m+1)$
 $x_2 = x_1 - 2 = -1 - 2 = -(1+2)$
 $x_3 = x_2 - 3 = -1 - 2 - 3$

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Question 10

A child was asked to add first few natural numbers (i.e. $1 + 2 + 3 + \dots$) so long his patience permitted. As he stopped, he gave the sum as 575. When the teacher declared the result wrong, the child discovered he had missed one number in the sequence during addition. The number he missed was

- A less than 10
- B 10
- C 15
- D more than 15

A child was asked to add first few natural numbers (i.e. $1 + 2 + 3 + \dots$) so long his patience permitted. As he stopped, he gave the sum as 575. When the teacher declared the result wrong, the child discovered he had missed one number in the sequence during addition. The number he missed was

A) less than 10
B) 10
C) 15
D) more than 15

$\frac{n(n+1)}{2} = 575$
 $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$
 $35 \times 36 = 1260$
 $21 \times 22 = 462$
 $21 \times 22 = 462$
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Instructions

DIRECTIONS for the following questions: These questions are based on the situation given below: There are fifty integers $a_1, a_2, \dots, a_{50}$, not all of them necessarily different. Let the greatest integer of these fifty integers be referred to as G , and the smallest integer be referred to as L . The integers a_1 through a_{24} form sequence $S1$, and the rest form sequence $S2$. Each member of $S1$ is less than or equal to each member of $S2$.

Question 11

Every element of S_1 is made greater than or equal to every element of S_2 by adding to each element of S_1 an integer x . Then x cannot be less than:

- A 2^{10}
- B The smallest value of S_2
- C The largest value of S_2
- D $(G-L)$

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Question 12

Elements of S_1 are in ascending order, and those of S_2 are in descending order. a_{24} and a_{25} are interchanged. Then, which of the following statements is true?

- A S_1 continues to be in ascending order
- B S_2 continues to be in descending order
- C S_1 continues to be in ascending order and S_2 in descending order.
- D None of the above

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Question 13

All values in S_1 are changed in sign, while those in S_2 remain unchanged. Which of the following statements is true?

- A Every member of S_1 is greater than or equal to every member of S_2 .
- B G is in S_1
- C If all numbers originally in S_1 and S_2 had the same sign, then after the change of sign, the largest number of S_1 and S_2 is in S_1 .
- D None of the above

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Instructions

For the following questions answer them individually

Question 14

A square is drawn by joining the midpoints of the sides of a given square. A third square is drawn inside the second square in the same way and this process is continued indefinitely. If a side of the first square is 8 cm, the sum of the areas of all the squares such formed (in sq.cm.) is

- A 128
- B 120

- C 96
- D None of these

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Question 15

Consider a function $f(k)$ defined for positive integers $k = 1, 2, \dots$; the function satisfies the condition $f(1) + f(2) + \dots = \frac{p}{p-1}$. Where p is fraction i.e. $0 < p < 1$. Then $f(k)$ is given by

- A $(-p)(p)^{(k-1)}$
- B $p(1-p)^{k-1}$
- C $(p(1-p))^{k-1}$
- D None of these

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Question 16

Let $a_1, a_2, \dots, a_{3n}$ be an arithmetic progression with $a_1 = 3$ and $a_2 = 7$. If $a_1 + a_2 + \dots + a_{3n} = 1830$, then what is the smallest positive integer m such that $m(a_1 + a_2 + \dots + a_n) > 1830$?

- A 8
- B 9
- C 10
- D 11

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Question 17

The value of the sum $7 \times 11 + 11 \times 15 + 15 \times 19 + \dots + 95 \times 99$ is



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Question 18

Let $a_1, a_2, \dots$ be integers such that

$$a_1 - a_2 + a_3 - a_4 + \dots + (-1)^{n-1} a_n = n, \text{ for all } n \geq 1.$$

Then $a_{51} + a_{52} + \dots + a_{1023}$ equals

- A 0
- B 1
- C 10
- D -1



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Question 19

The number of common terms in the two sequences: $15, 19, 23, 27, \dots, 415$ and $14, 19, 24, 29, \dots, 464$ is

- A 21
- B 20
- C 18
- D 19

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The number of common terms in the two sequences: 15, 19, 23, 27, ..., 415 and 14, 18, 22, 26, ..., 464 is

A) 21
B) 20
C) 18
D) 19

1) First common term
19

2) LCM(4, 5) = 20

19 + 20 = 39

39 + 20 = 59

59 + 20 = 79

79 + 20 = 99

99 + 20 = 119

119 + 20 = 139

139 + 20 = 159

159 + 20 = 179

179 + 20 = 199

199 + 20 = 219

219 + 20 = 239

239 + 20 = 259

259 + 20 = 279

279 + 20 = 299

299 + 20 = 319

319 + 20 = 339

339 + 20 = 359

359 + 20 = 379

379 + 20 = 399

399 + 20 = 419

419 + 20 = 439

439 + 20 = 459

459 + 20 = 479

479 + 20 = 499

499 + 20 = 519

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9239 + 20 = 9259

9259 + 20 = 9279

9279 + 20 = 9299

9299 + 20 = 9319

9319 + 20 = 9339

9339 + 20 = 9359

9359 + 20 = 9379

9379 + 20 = 9399

9399 + 20 = 9419

9419 + 20 = 9439

9439 + 20 = 9459

9459 + 20 = 9479

9479 + 20 = 9499

9499 + 20 = 9519

9519 + 20 = 9539

9539 + 20 = 9559

9559 + 20 = 9579

9579 + 20 = 9599

9599 + 20 = 9619

9619 + 20 = 9639

9639 + 20 = 9659

9659 + 20 = 9679

9679 + 20 = 9699

9699 + 20 = 9719

9719 + 20 = 9739

9739 + 20 = 9759

9759 + 20 = 9779

9779 + 20 = 9799

9799 + 20 = 9819

9819 + 20 = 9839

9839 + 20 = 9859

9859 + 20 = 9879

9879 + 20 = 9899

9899 + 20 = 9919

9919 + 20 = 9939

9939 + 20 = 9959

9959 + 20 = 9979

9979 + 20 = 9999

9999 + 20 = 10019

10019 + 20 = 10039

10039 + 20 = 10059

10059 + 20 = 10079

10079 + 20 = 10099

10099 + 20 = 10119

10119 + 20 = 10139

10139 + 20 = 10159

10159 + 20 = 10179

10179 + 20 = 10199

10199 + 20 = 10219

10219 + 20 = 10239

10239 + 20 = 10259

10259 + 20 = 10279

10279 + 20 = 10299

10299 + 20 = 10319

10319 + 20 = 10339

10339 + 20 = 10359

10359 + 20 = 10379

10379 + 20 = 10399

10399 + 20 = 10419

10419 + 20 = 10439

10439 + 20 = 10459

10459 + 20 = 10479

10479 + 20 = 10499

10499 + 20 = 10519

10519 + 20 = 10539

10539 + 20 = 10559

10559 + 20 = 10579

10579 + 20 = 10599

10599 + 20 = 10619

10619 + 20 = 10639

10639 + 20 = 10659

10659 + 20 = 10679

10679 + 20 = 10699

10699 + 20 = 10719

10719 + 20 = 10739

10739 + 20 = 10759

10759 + 20 = 10779

10779 + 20 = 10799

10799 + 20 = 10819

10819 + 20 = 10839

10839 + 20 = 10859

10859 + 20 = 10879

10879 + 20 = 10899

10899 + 20 = 10919

10919 + 20 = 10939

10939 + 20 = 10959

10959 + 20 = 10979

10979 + 20 = 10999

10999 + 20 = 11019

11019 + 20 = 11039

11039 + 20 = 11059

11059 + 20 = 11079

11079 + 20 = 11099

11099 + 20 = 11119

11119 + 20 = 11139

11139 + 20 = 11159

11159 + 20 = 11179

11179 + 20 = 11199

11199 + 20 = 11219

11219 + 20 = 11239

11239 + 20 = 11259

11259 + 20 = 11279

11279 + 20 = 11299

11299 + 20 = 11319

11319 + 20 = 11339

11339 + 20 = 11359

11359 + 20 = 11379

11379 + 20 = 11399

11399 + 20 = 11419

11419 + 20 = 11439

11439 + 20 = 11459

11459 + 20 = 11479

11479 + 20 = 11499

11499 + 20 = 11519

11519 + 20 = 11539

11539 + 20 = 11559

11559 + 20 = 11579

11579 + 20 = 11599

11599 + 20 = 11619

11619 + 20 = 11639

11639 + 20 = 11659

11659 + 20 = 11679

11679 + 20 = 11699

11699 + 20 = 11719

11719 + 20 = 11739

11739 + 20 = 11759

11759 + 20 = 11779

11779 + 20 = 11799

11799 + 20 = 11819

11819 + 20 = 11839

11839 + 20 = 11859

11859 + 20 = 11879

11879 + 20 = 11899

11899 + 20 = 11919

11919 + 20 = 11939

11939 + 20 = 11959

11959 + 20 = 11979

11979 + 20 = 11999

11999 + 20 = 12019

12019 + 20 = 12039

12039 + 20 = 12059

12059 + 20 = 12079

12079 + 20 = 12099

12099 + 20 = 12119

12119 + 20 = 12139

A lab experiment measures the number of organisms at 8 am every day. Starting with 2 organisms on the first day, the number of organisms on any day is equal to 3 more than twice the number on the previous day. If the number of organisms on the n th day exceeds one million, then the lowest possible value of n is

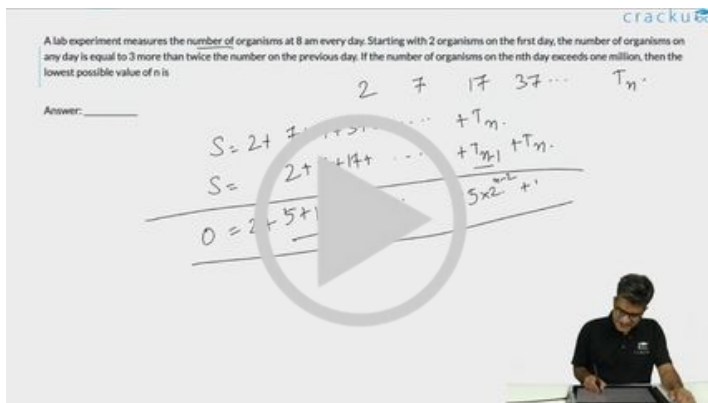
Answer: _____

Handwritten solution:

$$2 \quad 7 \quad 17 \quad 37 \dots T_n$$

$$S = 2 + 7 + 17 + 37 + \dots + T_n$$

$$S = 2 + 7 + 17 + 37 + \dots + T_{n-1} + T_n$$

$$0 = 2 + 5 + 10 + 15 + \dots + 5 \times 2 + 1$$


VIDEO SOLUTION



CAT FORMULAS BOOK

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Question 22

Let a_n and b_n be two sequences such that $a_n = 13 + 6(n - 1)$ and $b_n = 15 + 7(n - 1)$ for all natural numbers n . Then, the largest three digit integer that is common to both these sequences, is

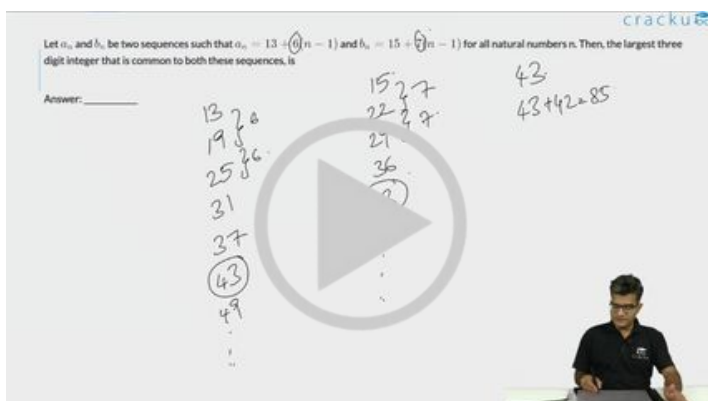
Let a_n and b_n be two sequences such that $a_n = 13 + 6(n - 1)$ and $b_n = 15 + 7(n - 1)$ for all natural numbers n . Then, the largest three digit integer that is common to both these sequences, is

Answer: _____

Handwritten solution:

$$13, 19, 25, 31, 37, 43, 49, \dots$$

$$15, 22, 29, 36, 43, 50, \dots$$

$$43 + 42 = 85$$


VIDEO SOLUTION

Question 23

The arithmetic mean of x, y and z is 80, and that of x, y, z, u and v is 75, where $u = (x+y)/2$ and $v = (y+z)/2$. If $x \geq z$, then the minimum possible value of x is

The arithmetic mean of x, y and z is 80, and that of x, y, z, u and v is 75, where $u = (x+y)/2$ and $v = (y+z)/2$. If $x \geq z$, then the minimum possible value of x is

Handwritten solution:

$$\frac{x+y+z}{3} = 80 \Rightarrow x+y+z = 240$$

$$\frac{x+y+z+u+v}{5} = 75 \Rightarrow x+y+z+u+v = 375$$

$$u = \frac{x+y}{2}, v = \frac{y+z}{2}$$

$$x+y+z + \frac{x+y}{2} + \frac{y+z}{2} = 375$$

$$\frac{2x+2y+2z+x+y+y+z}{2} = 375$$

$$\frac{3x+3y+3z}{2} = 375$$

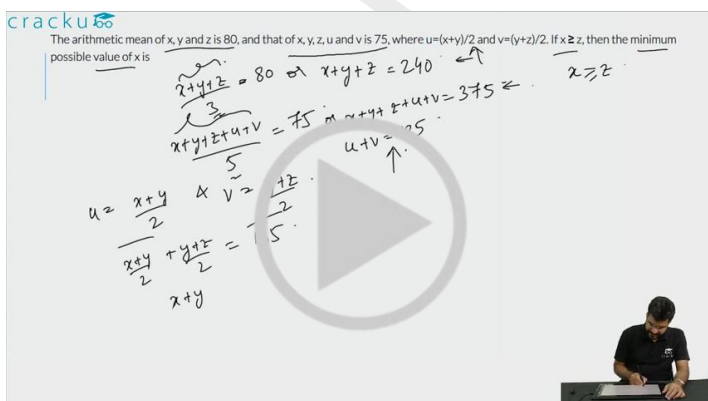
$$3(x+y+z) = 750$$

$$x+y+z = 250$$

$$240 = 250 - z$$

$$z = 10$$

$$x+y = 230$$

$$x \geq z \Rightarrow x \geq 10$$


Question 24

Let $x = \sqrt{4 + \sqrt{4 - \sqrt{4 + \sqrt{4 - \dots}}}}$ to infinity. Then x equals

- A 3
- B $(\sqrt{13} - 1)/2$
- C $(\sqrt{13} + 1)/2$
- D $\sqrt{13}$

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Question 25

Let x, y, z be three positive real numbers in a geometric progression such that $x < y < z$. If 5x, 16y, and 12z are in an arithmetic progression then the common ratio of the geometric progression is

- A $\frac{3}{6}$
- B $\frac{1}{6}$
- C $\frac{5}{2}$
- D $\frac{3}{2}$

cracku

Let x, y, z be three positive real numbers in a geometric progression such that $x < y < z$. If 5x, 16y, and 12z are in an arithmetic progression then the common ratio of the geometric progression is

Handwritten solution:

Let x, y, z be in GP with common ratio r. $r > 1$.

x, y, z are in GP.

$x \cdot r = y$, $y \cdot r = z$.

$5x + 12z = 2 \cdot 16y$.

$5x + 12x \cdot r^2 = 32xy$.

$5 + 12r^2 = 32r$.

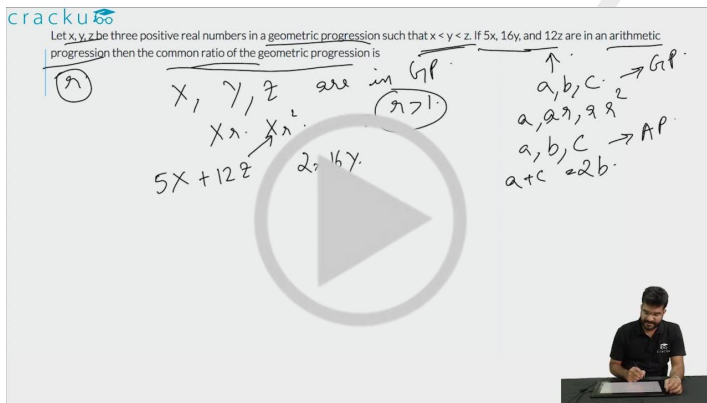
$12r^2 - 32r + 5 = 0$.

$(4r - 5)(3r - 1) = 0$.

$r = \frac{5}{4}$ or $r = \frac{1}{3}$.

Since $r > 1$, $r = \frac{5}{4}$.

Answer: C



Question 26

The number of common terms in the two sequences 17, 21, 25, ..., 417 and 16, 21, 26, ..., 466 is

- A 78
- B 19
- C 20

D 77

E 22

The number of common terms in the two sequences $17, 21, 25, \dots, 417$ and $16, 21, 26, \dots, 466$ is

A) 78
B) 19
C) 20
D) 77

$a = 21$
 $d = \text{LCM}(4, 5) = 20$



 VIDEO SOLUTION

Question 27

The 288th term of the series $a, b, b, c, c, c, d, d, d, d, e, e, e, e, f, f, f, f, \dots$ is

A u

B v

C w

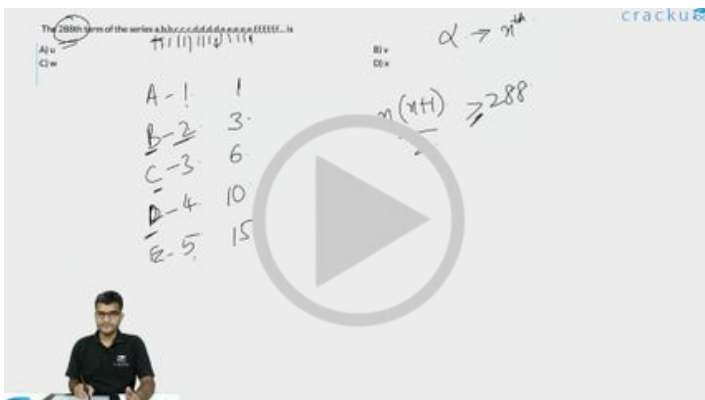
D x

The 288th term of the series $a, b, b, c, c, c, d, d, d, d, e, e, e, e, f, f, f, f, \dots$ is

A) u
B) v
C) w
D) x

$\alpha \rightarrow x$
 $n(x+1) \geq 288$

A-1	1
B-2	3
C-3	6
D-4	10
E-5	15



 VIDEO SOLUTION

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Question 28

In a certain examination paper, there are n questions. For $j = 1, 2, \dots, n$, there are 2^{n-j} students who answered j or more questions wrongly. If the total number of wrong answers is 4095, then the value of n is

A 12

B 11

C 10

D 9

[VIEW SOLUTION](#)

Question 29

A group of 630 children is arranged in rows for a group photograph session. Each row contains three fewer children than the row in front of it. What number of rows is not possible?

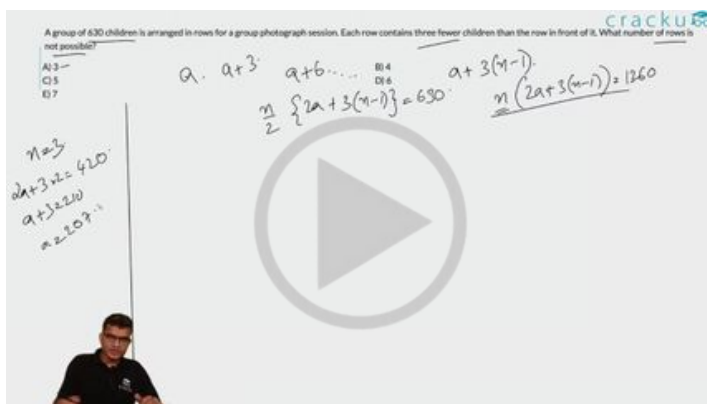
A 3

B 4

C 5

D 6

E 7



[VIDEO SOLUTION](#)

Question 30

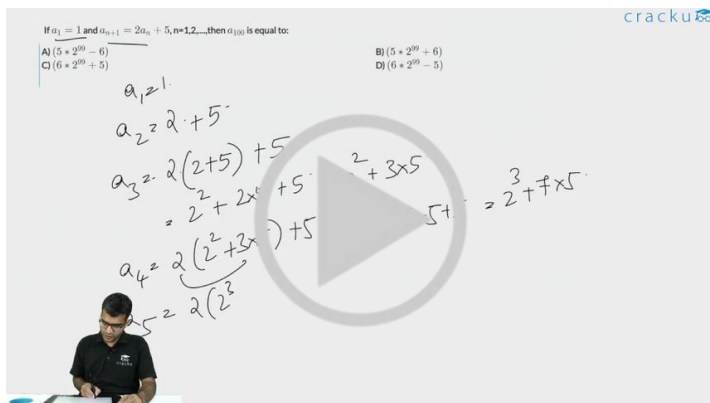
If $a_1 = 1$ and $a_{n+1} = 2a_n + 5$, $n=1,2,\dots$, then a_{100} is equal to:

A $(5 * 2^{99} - 6)$

B $(5 * 2^{99} + 6)$

C $(6 * 2^{99} + 5)$

D $(6 * 2^{99} - 5)$



VIDEO SOLUTION

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Question 31

The n th element of a series is represented as

$$X_n = (-1)^n X_{n-1}$$

If $X_0 = x$ and $x > 0$, then which of the following is always true?

- A X_n is positive if n is even
- B X_n is positive if n is odd
- C X_n is negative if n is even
- D None of these

VIEW SOLUTION

Question 32

Let S denotes the infinite sum $2 + 5x + 9x^2 + 14x^3 + 20x^4 + \dots$, where $|x| < 1$ and the coefficient of x^{n-1} is $n(n+3)/2$, ($n = 1, 2, \dots$). Then S equals:

- A $(2 - x)/(1 - x)^3$
- B $(2 - x)/(1 + x)^3$
- C $(2 + x)/(1 - x)^3$
- D $(2 + x)/(1 + x)^3$

VIEW SOLUTION

Question 33

Consider a sequence of real numbers, $x_1, x_2, x_3, \dots$ such that $x_{n+1} = x_n + n - 1$ for all $n \geq 1$. If $x_1 = -1$ then x_{100} is equal to

- A 4849
- B 4949

C 4950

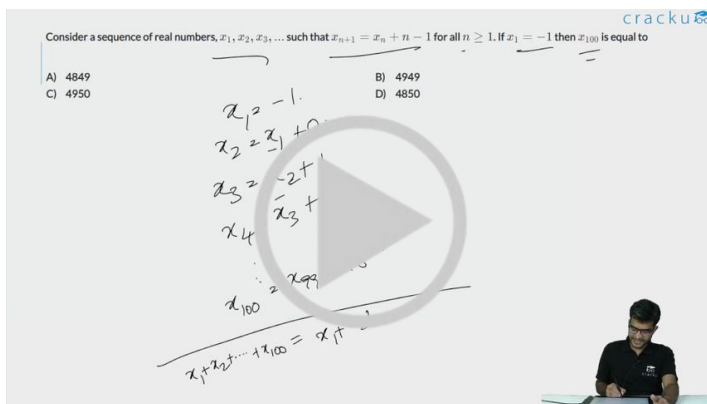
D 4850

cracku

Consider a sequence of real numbers, $x_1, x_2, x_3, \dots$ such that $x_{n+1} = x_n + n - 1$ for all $n \geq 1$. If $x_1 = -1$ then x_{100} is equal to

A) 4849
B) 4949
C) 4950
D) 4850

$x_1 = -1$
 $x_2 = x_1 + 0$
 $x_3 = x_2 + 1$
 $x_4 = x_3 + 2$
 $\vdots$
 $x_{100} = x_{99} + 99$
 $x_1 + x_2 + \dots + x_{100} = x_1 + \dots$



VIDEO SOLUTION

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Question 34

If $a_1 = \frac{1}{2 \times 5}$, $a_2 = \frac{1}{5 \times 8}$, $a_3 = \frac{1}{8 \times 11}$, ..., then $a_1 + a_2 + a_3 + \dots + a_{100}$ is

A $\frac{25}{151}$

B $\frac{1}{2}$

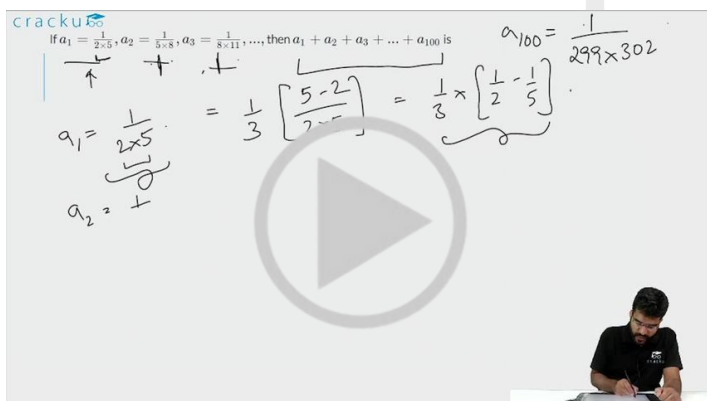
C $\frac{1}{4}$

D $\frac{111}{55}$

cracku

If $a_1 = \frac{1}{2 \times 5}$, $a_2 = \frac{1}{5 \times 8}$, $a_3 = \frac{1}{8 \times 11}$, ..., then $a_1 + a_2 + a_3 + \dots + a_{100}$ is

$a_1 = \frac{1}{2 \times 5}$
 $a_2 = \frac{1}{5 \times 8}$
 $a_3 = \frac{1}{8 \times 11}$
 $\vdots$
 $a_{100} = \frac{1}{299 \times 302}$
 $\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \dots + \frac{1}{299 \times 302}$
 $= \frac{1}{3} \left[\frac{5-2}{2 \times 5} \right] = \frac{1}{3} \times \left[\frac{1}{2} - \frac{1}{5} \right]$



VIDEO SOLUTION

Question 35

An infinite geometric progression $a_1, a_2, \dots$ has the property that $a_n = 3(a_{n+1} + a_{n+2} + \dots)$ for every $n \geq 1$.

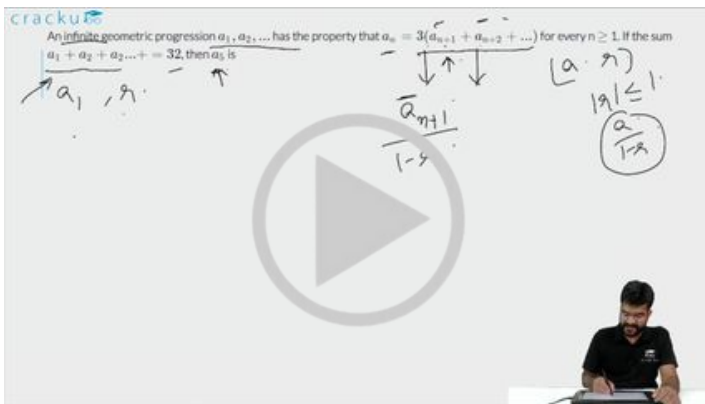
If the sum $a_1 + a_2 + a_3 + \dots = 32$, then a_5 is

A $1/32$

B $2/32$

C 3/32

D 4/32



VIDEO SOLUTION

Question 36

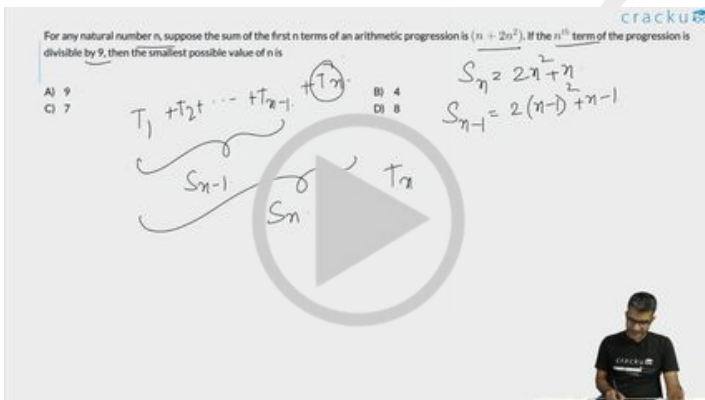
For any natural number n , suppose the sum of the first n terms of an arithmetic progression is $(n + 2n^2)$. If the n^{th} term of the progression is divisible by 9, then the smallest possible value of n is

A 9

B 4

C 7

D 8



VIDEO SOLUTION

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Question 37

Consider the arithmetic progression 3, 7, 11, ... and let A_n denote the sum of the first n terms of this progression. Then the value of $\frac{1}{25} \sum_{n=1}^{25} A_n$ is

A 455

- B 442
- C 415
- D 404

cracku

Consider the arithmetic progression 3, 7, 11, ... and let A_n denote the sum of the first n terms of this progression. Then the value of $\frac{1}{25} \sum_{n=1}^{25} A_n$ is

A) 455
C) 415

B) 442
D) 404

$a = 3$
 $d = 4$
 $A_n = \frac{n}{2} \{2a + (n-1)d\} = \frac{n}{2} \{2 \cdot 3 + (n-1) \cdot 4\} = \frac{n}{2} \{4n + 2\} = \frac{n}{2} \{2(2n + 1)\} = n(2n + 1)$
 $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$
 $\sum_{n=1}^{25} A_n = \sum_{n=1}^{25} n(2n + 1) = 2 \sum_{n=1}^{25} n^2 + \sum_{n=1}^{25} n$

VIDEO SOLUTION

Question 38

For a real number x , if $\frac{1}{2}$, $\frac{\log_3(2^x - 9)}{\log_3 4}$, and $\frac{\log_5(2^x + \frac{17}{2})}{\log_5 4}$ are in an arithmetic progression, then the common difference is

- A $\log_4 \left(\frac{3}{2}\right)$
- B $\log_4 7$
- C $\log_4 \left(\frac{23}{2}\right)$
- D $\log_4 \left(\frac{7}{2}\right)$

cracku

For a real number x , if $\frac{1}{2}$, $\frac{\log_3(2^x - 9)}{\log_3 4}$, and $\frac{\log_5(2^x + \frac{17}{2})}{\log_5 4}$ are in an arithmetic progression, then the common difference is

A) $\log_4 \left(\frac{3}{2}\right)$
C) $\log_4 \left(\frac{23}{2}\right)$

B) $\log_4 7$
D) $\log_4 \left(\frac{7}{2}\right)$

$\log_4 2, \log_4(2^x - 9), \log_4(2^x + \frac{17}{2})$
 $\log_4 2 + \log_4 2^2 = 1 + 2 = 3$
 $\log_4 2 + \log_4 2^2 = 1 + 2 = 3$
 $\frac{1}{2} = \log_4 2$

VIDEO SOLUTION

Question 39

The value of $1 + \left(1 + \frac{1}{3}\right) \frac{1}{4} + \left(1 + \frac{1}{3} + \frac{1}{9}\right) \frac{1}{16} + \left(1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27}\right) \frac{1}{64} + \dots$ is

- A $\frac{15}{13}$
- B $\frac{27}{12}$

C $\frac{15}{8}$

D $\frac{16}{11}$

The value of $1 + \left(\frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots\right) + \left(\frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \dots\right) + \dots$ is

A) $\frac{27}{16}$
B) $\frac{16}{11}$
C) $\frac{15}{8}$
D) $\frac{16}{11}$

$a, ar, ar^2, \dots \infty$
 $-1 < r < 1$
 $\frac{a}{1-r}$

$1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots$
 $= \frac{1}{1 - \frac{1}{4}}$
 $= \frac{4}{3}$

$\frac{1}{3} \left[\frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots \right]$
 $= \frac{1}{3} \times \frac{1}{4} \left[1 + \frac{1}{4} + \frac{1}{16} + \dots \right]$

VIDEO SOLUTION

CAT Percentile Predictor

Question 40

Let $a_n = 46 + 8n$ and $b_n = 98 + 4n$ be two sequences for natural numbers $n \leq 100$. Then, the sum of all terms common to both the sequences is

- A 14900
B 14798
C 15000
D 14602

Let $a_n = 46 + 8n$ and $b_n = 98 + 4n$ be two sequences for natural numbers $n \leq 100$. Then, the sum of all terms common to both the sequences is

A) 14900
B) 14798
C) 15000
D) 14602

$54, 62, 70, 78, 86, 94, 102, 110, 118, 126, 134, 142, 150, 158, 166, 174, 182, 190, 198, 206, 214, 222, 230, 238, 246, 254, 262, 270, 278, 286, 294, 302, 310, 318, 326, 334, 342, 350, 358, 366, 374, 382, 390, 398, 406, 414, 422, 430, 438, 446, 454, 462, 470, 478, 486, 494, 502, 510, 518, 526, 534, 542, 550, 558, 566, 574, 582, 590, 598, 606, 614, 622, 630, 638, 646, 654, 662, 670, 678, 686, 694, 702, 710, 718, 726, 734, 742, 750, 758, 766, 774, 782, 790, 798, 806, 814, 822, 830, 838, 846, 854, 862, 870, 878, 886, 894, 902, 910, 918, 926, 934, 942, 950, 958, 966, 974, 982, 990, 998$

$102, 106, 110, 114, 118, 122, 126, 130, 134, 138, 142, 146, 150, 154, 158, 162, 166, 170, 174, 178, 182, 186, 190, 194, 198, 202, 206, 210, 214, 218, 222, 226, 230, 234, 238, 242, 246, 250, 254, 258, 262, 266, 270, 274, 278, 282, 286, 290, 294, 298, 302, 306, 310, 314, 318, 322, 326, 330, 334, 338, 342, 346, 350, 354, 358, 362, 366, 370, 374, 378, 382, 386, 390, 394, 398, 402, 406, 410, 414, 418, 422, 426, 430, 434, 438, 442, 446, 450, 454, 458, 462, 466, 470, 474, 478, 482, 486, 490, 494, 498, 502, 506, 510, 514, 518, 522, 526, 530, 534, 538, 542, 546, 550, 554, 558, 562, 566, 570, 574, 578, 582, 586, 590, 594, 598, 602, 606, 610, 614, 618, 622, 626, 630, 634, 638, 642, 646, 650, 654, 658, 662, 666, 670, 674, 678, 682, 686, 690, 694, 698, 702, 706, 710, 714, 718, 722, 726, 730, 734, 738, 742, 746, 750, 754, 758, 762, 766, 770, 774, 778, 782, 786, 790, 794, 798, 802, 806, 810, 814, 818, 822, 826, 830, 834, 838, 842, 846, 850, 854, 858, 862, 866, 870, 874, 878, 882, 886, 890, 894, 898, 902, 906, 910, 914, 918, 922, 926, 930, 934, 938, 942, 946, 950, 954, 958, 962, 966, 970, 974, 978, 982, 986, 990, 994, 998$

$102 + 110 + 118 + \dots$

VIDEO SOLUTION

Question 41

Three times the first of three consecutive odd integers is 3 more than twice the third. What is the third integer?

- A 15
B 9

C 11

D 5

[VIEW SOLUTION](#)

Question 42

If $a_1 + a_2 + a_3 + \dots + a_n = 3(2^{n+1} - 2)$, for every $n \geq 1$, then a_{11} equals

cracku

If $a_1 + a_2 + a_3 + \dots + a_n = 3(2^{n+1} - 2)$, for every $n \geq 1$, then a_{11} equals

$$\begin{aligned} \text{① } a_1 + a_2 + \dots + a_{11} &= 3(2^{12} - 2) \quad n=11 \\ a_1 + a_2 + \dots + a_{10} &= 3(2^{11} - 2) \quad n=10 \end{aligned}$$



[VIDEO SOLUTION](#)

Important Verbal Ability Questions for CAT (Download PDF)

Question 43

Suppose $x_1, x_2, x_3, \dots, x_{100}$ are in arithmetic progression such that $x_5 = -4$ and $2x_6 + 2x_9 = x_{11} + x_{13}$. Then, x_{100} equals

A -194

B -196

C 204

D 206

cracku

Suppose $x_1, x_2, x_3, \dots, x_{100}$ are in arithmetic progression such that $x_5 = -4$ and $2x_6 + 2x_9 = x_{11} + x_{13}$. Then, x_{100} equals

A) -194
B) -196
C) 204
D) 206

$$\begin{aligned} x_5 &= -4 + d \\ x_6 &= -4 + 2d \\ x_9 &= -4 + 5d \\ x_{11} &= -4 + 7d \\ x_{13} &= -4 + 9d \end{aligned}$$

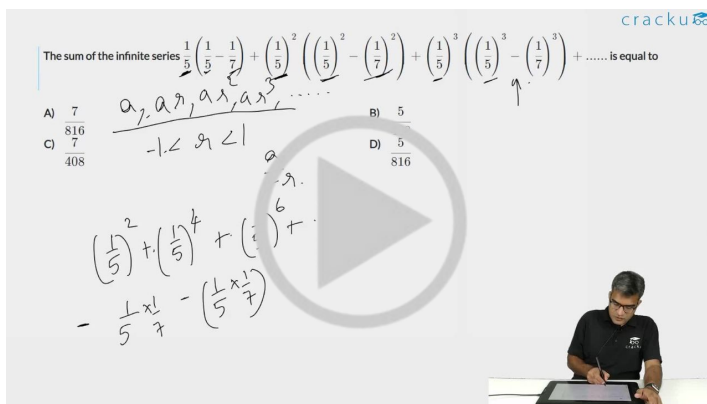


[VIDEO SOLUTION](#)

Question 44

The sum of the infinite series $\frac{1}{5} \left(\frac{1}{5} - \frac{1}{7} \right) + \left(\frac{1}{5} \right)^2 \left(\left(\frac{1}{5} \right)^2 - \left(\frac{1}{7} \right)^2 \right) + \left(\frac{1}{5} \right)^3 \left(\left(\frac{1}{5} \right)^3 - \left(\frac{1}{7} \right)^3 \right) + \dots$ is equal to

- A $\frac{7}{816}$
- B $\frac{5}{408}$
- C $\frac{7}{408}$
- D $\frac{5}{816}$



[VIDEO SOLUTION](#)

Question 45

The sum of 3rd and 15th elements of an arithmetic progression is equal to the sum of 6th, 11th and 13th elements of the same progression. Then which element of the series should necessarily be equal to zero?

- A 1st
- B 9th
- C 12th
- D None of the above

The sum of 3rd and 15th elements of an arithmetic progression is equal to the sum of 6th, 11th and 13th elements of the same progression. Then which element of the series should necessarily be equal to zero?

A) 1st
B) 9th
C) 12th
D) None of the above

a, d
 $T_n = a + (n-1)d$
 $S_n = \frac{n}{2} [2a + (n-1)d]$

$a + 2d + a + 14d$

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Data Interpretation for CAT Questions (download pdf)

Question 46

Consider the sequence $t_1 = 1, t_2 = -1$ and $t_n = \left(\frac{n-3}{n-1}\right) t_{n-2}$ for $n \geq 3$. Then, the value of the sum

$$\frac{1}{t_2} + \frac{1}{t_4} + \frac{1}{t_6} + \dots + \frac{1}{t_{2022}} + \frac{1}{t_{2024}}, \text{ is}$$

- A -1024144
- B -1022121
- C -1023132
- D -1026169

Consider the sequence $t_1 = 1, t_2 = -1$ and $t_n = \left(\frac{n-3}{n-1}\right) t_{n-2}$ for $n \geq 3$. Then, the value of the sum $\frac{1}{t_2} + \frac{1}{t_4} + \frac{1}{t_6} + \dots + \frac{1}{t_{2022}} + \frac{1}{t_{2024}}$, is

A) -1024144
B) -1022121
C) -1023132
D) -1026169

$t_1 = 1$
 $t_2 = -1$
 $t_3 = \frac{0}{2} \times t_1 = 0$
 $t_4 = \frac{1}{3} \times t_2 = -\frac{1}{3}$
 $t_5 = 0$
 $t_6 = \frac{3}{5} \times t_4 = -\frac{3}{5}$
 $t_7 = 0$

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Instructions

Answer the questions based on the following information. A series S_1 of five positive integers is such that the third term is half the first term and the fifth term is 20 more than the first term. In series S_2 , the n th term is defined as the difference between the $(n+1)$ th term and the n th term of series S_1 . S_2 is an arithmetic progression with a common difference of 30.

Question 47

What is the average value of the terms of series S_1 ?

- A 60
- B 70
- C 80
- D Average is not an integer

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Instructions

For the following questions answer them individually

Question 48

What is the value of the following expression?

$$(1/(2^2 - 1)) + (1/(4^2 - 1)) + (1/(6^2 - 1)) + \dots + (1/(20^2 - 1))$$

- A 9/19
- B 10/19
- C 10/21
- D 11/21

What is the value of the following expression? $(1/(2^2 - 1)) + (1/(4^2 - 1)) + (1/(6^2 - 1)) + \dots + (1/(20^2 - 1))$

A) 9/19
B) 10/19
C) 10/21
D) 11/21

$x^2 - 1 = (x-1)(x+1)$

$S = \frac{1}{2^2-1} + \frac{1}{4^2-1} + \frac{1}{6^2-1} + \dots + \frac{1}{20^2-1}$

$= \frac{1}{(2-1)(2+1)} + \frac{1}{(4-1)(4+1)} + \frac{1}{(6-1)(6+1)} + \dots + \frac{1}{(20-1)(20+1)}$

$= \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{19 \times 21}$

[VIDEO SOLUTION](#)

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Question 49

If the square of the 7th term of an arithmetic progression with positive common difference equals the product of the 3rd and 17th terms, then the ratio of the first term to the common difference is

- A 2:3
- B 3:2
- C 3:4
- D 4:3

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If the square of the 7th term of an arithmetic progression with positive common difference equals the product of the 3rd and 17th terms, then the ratio of the first term to the common difference is

A) 2:3
B) 3:2
C) 3:4
D) 4:3

a
 $d > 0$
 7th term $= a + 6d$
 3rd term $= a + 2d$
 17th term $= a + 16d$



VIDEO SOLUTION

Question 50

If $(2n + 1) + (2n + 3) + (2n + 5) + \dots + (2n + 47) = 5280$, then what is the value of $1 + 2 + 3 + \dots + n$?

cracku

If $(2n + 1) + (2n + 3) + (2n + 5) + \dots + (2n + 47) = 5280$, then what is the value of $1 + 2 + 3 + \dots + n$?

$n = ?$
 $1, 3, 5, \dots$
 $\frac{47 - 1}{2} + 1 = 23 + 1 = 24$ Terms
 $2n \times 24 + (1 + 3 + \dots)$
 $= 5280$



VIDEO SOLUTION

Question 51

The natural numbers are divided into groups as (1), (2, 3, 4), (5, 6, 7, 8, 9), and so on. Then, the sum of the numbers in the 15th group is equal to

- A 6119
- B 6090
- C 4941
- D 7471

The natural numbers are divided into groups as (1), (2, 3, 4), (5, 6, 7, 8, 9), and so on. Then, the sum of the numbers in the 15th group is equal to

A) 6119
B) 6090
C) 4941
D) 7471

Handwritten notes:
Last No. 15th group = 225
Last No. 14th group = 196
17th 15th ...

VIDEO SOLUTION

Quantitative Aptitude for CAT Questions (download pdf)

Question 52

Let $a_1, a_2, \dots, a_{52}$ be positive integers such that $a_1 < a_2 < \dots < a_{52}$. Suppose, their arithmetic mean is one less than arithmetic mean of $a_2, a_3, \dots, a_{52}$. If $a_{52} = 100$, then the largest possible value of a_1 is

- A 48
- B 20
- C 23
- D 45

Let $a_1, a_2, \dots, a_{52}$ be positive integers such that $a_1 < a_2 < \dots < a_{52}$. Suppose, their arithmetic mean is one less than arithmetic mean of $a_2, a_3, \dots, a_{52}$. If $a_{52} = 100$, then the largest possible value of a_1 is

Handwritten notes:
 $a_1 + a_2 + \dots + a_{52} = A - 1$
 $a_2 + a_3 + \dots + a_{52} = A$
 $a_1 = A - 1 - (A - a_1) = 2a_1 - 1$
 $a_1 = 48$

VIDEO SOLUTION

Question 53

Consider a sequence where the n^{th} term, $t_n = n/(n+2)$, $n = 1, 2, \dots$. The value of $t_3 * t_4 * t_5 * \dots * t_{53}$ equals.

- A $2/495$
- B $2/477$
- C $12/55$
- D $1/1485$

E 1/2970

Consider a sequence where the n^{th} term, $t_n = n/(n+2)$, $n = 1, 2, \dots$. The value of $t_3 + t_4 + \dots + t_{53}$ equals.

A) 2/495
B) 2/477
C) 12/55
D) 1/1485
E) 1/2970

Handwritten solution showing the sequence terms $t_3 = \frac{3}{5}$, $t_4 = \frac{4}{6}$, $t_5 = \frac{5}{7}$, $t_6 = \frac{6}{8}$, ..., $t_{53} = \frac{53}{55}$. The sum is calculated as $\frac{3}{5} \times \frac{5}{6} \times \frac{6}{7} \times \frac{7}{8} \times \dots \times \frac{53}{55} = \frac{3}{55}$.

VIDEO SOLUTION

Question 54

Three positive integers x , y and z are in arithmetic progression. If $y - x > 2$ and $xyz = 5(x + y + z)$, then $z - x$ equals

- A 8
- B 12
- C 14
- D 10

Three positive integers x , y and z are in arithmetic progression. If $y - x > 2$ and $xyz = 5(x + y + z)$, then $z - x$ equals

Handwritten solution showing the sequence terms x, y, z and their sum $x + y + z = 3x + 3d$. The equation $xyz = 5(x + y + z)$ is used to find $z - x$.

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Question 55

For a Fibonacci sequence, from the third term onwards, each term in the sequence is the sum of the previous two terms in that sequence. If the difference in squares of 7th and 6th terms of this sequence is 517, what is the 10th term of this sequence?

It is known that all the terms of the Fibonacci sequence are natural numbers

- A 147

- B 76
- C 123
- D Cannot be determined

cracku

For a Fibonacci sequence, from the third term onwards, each term in the sequence is the sum of the previous two terms in that sequence. If the difference in squares of 7th and 6th terms of this sequence is 517, what is the 10th term of this sequence?

It is known that all the terms of the Fibonacci sequence are natural numbers

A) 147
B) 76
C) 123
D) Cannot be determined

VIDEO SOLUTION

Question 56

Let $t_1, t_2, \dots$ be real numbers such that $t_1 + t_2 + \dots + t_n = 2n^2 + 9n + 13$, for every positive integer $n \geq 2$. If $t_k = 103$, then k equals

cracku

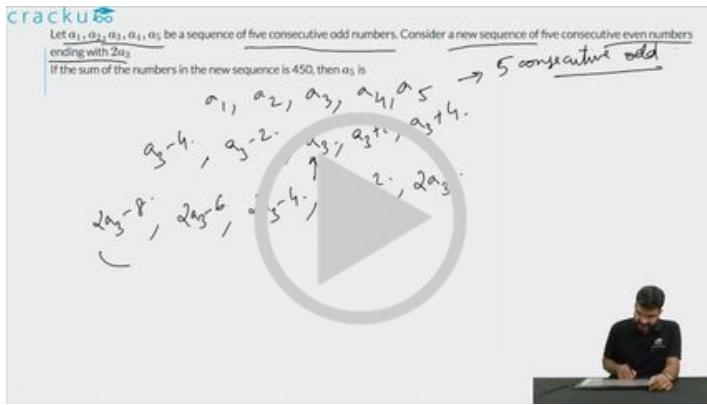
Let $t_1, t_2, \dots$ be real numbers such that $t_1 + t_2 + \dots + t_n = 2n^2 + 9n + 13$, for every positive integer $n \geq 2$. If $t_k = 103$, then k equals

VIDEO SOLUTION

Question 57

Let a_1, a_2, a_3, a_4, a_5 be a sequence of five consecutive odd numbers. Consider a new sequence of five consecutive even numbers ending with $2a_3$

If the sum of the numbers in the new sequence is 450, then a_5 is



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Question 58

On January 1, 2004 two new societies S1 and S2 are formed, each n numbers. On the first day of each subsequent month, S1 adds b members while S2 multiplies its current numbers by a constant factor r . Both the societies have the same number of members on July 2, 2004. If $b = 10.5n$, what is the value of r ?

- A 2.0
- B 1.9
- C 1.8
- D 1.7

VIDEO SOLUTION

Question 59

Two men X and Y started working for a certain company at similar jobs on January 1, 1950. X asked for an initial monthly salary of Rs. 300 with an annual increment of Rs. 30. Y asked for an initial monthly salary of Rs. 200 with a rise of Rs. 15 every 6 months. Assume that the arrangements remained unaltered till December 31, 1959. Salary is paid on the last day of the month. What is the total amount paid to them as salary during the period?

- A Rs. 93,300
- B Rs. 93,200
- C Rs. 93,100
- D None of these

VIDEO SOLUTION

Question 60

There are 8436 steel balls, each with a radius of 1 centimeter, stacked in a pile, with 1 ball on top, 3 balls in the second layer, 6 in the third layer, 10 in the fourth, and so on. The number of horizontal layers in the pile is

- A 34

- B 38
- C 36
- D 32

There are 8436 steel balls, each with a radius of 1 centimeter, stacked in a pile, with 1 ball on top, 3 balls in the second layer, 6 in the third layer, 10 in the fourth, and so on. The number of horizontal layers in the pile is

A) 34
B) 38
C) 36
D) 32

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Question 61

If the product of n positive real numbers is unity, then their sum is necessarily

- A a multiple of n
- B equal to $n + 1/n$
- C never less than n
- D a positive integer

If the product of n positive real numbers is unity, then their sum is necessarily

A) a multiple of n
B) equal to $n + 1/n$
C) never less than n
D) a positive integer

Answer: C

VIDEO SOLUTION

Question 62

N the set of natural numbers is partitioned into subsets $S_1 = (1)$, $S_2 = (2, 3)$, $S_3 = (4, 5, 6)$, $S_4 = (7, 8, 9, 10)$ and so on. The sum of the elements of the subset S_{50} is

- A 61250

- B 65525
- C 42455
- D 62525

cracku

The set of natural numbers is partitioned into subsets $S_1 = \{1\}$, $S_2 = \{2, 3\}$, $S_3 = \{4, 5, 6\}$, $S_4 = \{7, 8, 9, 10\}$ and so on. The sum of the elements of the subset S_{50} is

A) 61250
C) 42455

B) 65525
D) 62525

Let's Number $S_n = \frac{n(n+1)}{2}$

$S_{50} = \frac{50 \times 51}{2}$
 $= 51 \times 25 = 1275$

$S_{49} = \frac{49 \times 50}{2} = 49$

VIDEO SOLUTION

Question 63

What is the sum of the following series: $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} \dots + \frac{1}{100 \times 101}$?

- A $\frac{99}{100}$
- B $\frac{1}{100}$
- C $\frac{100}{101}$
- D $\frac{101}{102}$

cracku

What is the sum of the following series: $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} \dots + \frac{1}{100 \times 101}$?

A) $\frac{99}{100}$
C) $\frac{100}{101}$

B) $\frac{1}{100}$
D) $\frac{101}{102}$

$\frac{1}{1 \times 2} = \frac{1}{1} - \frac{1}{2}$

$\frac{1}{2 \times 3} = \frac{1}{2} - \frac{1}{3}$

$\frac{1}{3 \times 4} = \frac{1}{3} - \frac{1}{4}$

$\dots$

$\frac{1}{100 \times 101} = \frac{1}{100} - \frac{1}{101}$

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Instructions

Answer the questions based on the following information. A series S_1 of five positive integers is such that the third term is half the first term and the fifth term is 20 more than the first term. In series S_2 , the n th term is defined as the difference between the $(n+1)$ th term and the n th term of series S_1 , S_2 is an arithmetic progression with a common difference of 30.

Question 64

What is the sum of series S_2 ?

- A 10
- B 20
- C 30
- D 40

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Question 65

Fourth term of S_2

- A 50
- B 60
- C 70
- D None of these

[VIEW SOLUTION](#)

Instructions

For the following questions answer them individually

Question 66

If the sum of the first 11 terms of an arithmetic progression equals that of the first 19 terms, then what is the sum of the first 30 terms?

- A 0
- B -1
- C 1
- D Not unique

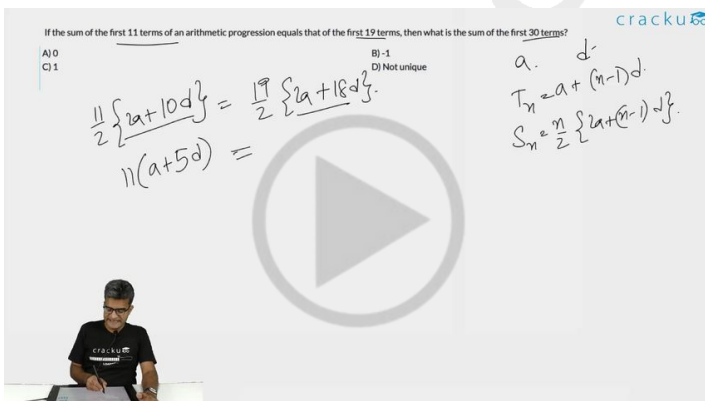
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If the sum of the first 11 terms of an arithmetic progression equals that of the first 19 terms, then what is the sum of the first 30 terms?

A) 0
B) -1
C) 1
D) Not unique

$$\frac{11}{2} \{2a + 10d\} = \frac{19}{2} \{2a + 18d\}$$
$$11(a + 5d) = 19(a + 9d)$$

$$T_n = a + (n-1)d$$
$$S_n = \frac{n}{2} \{2a + (n-1)d\}$$

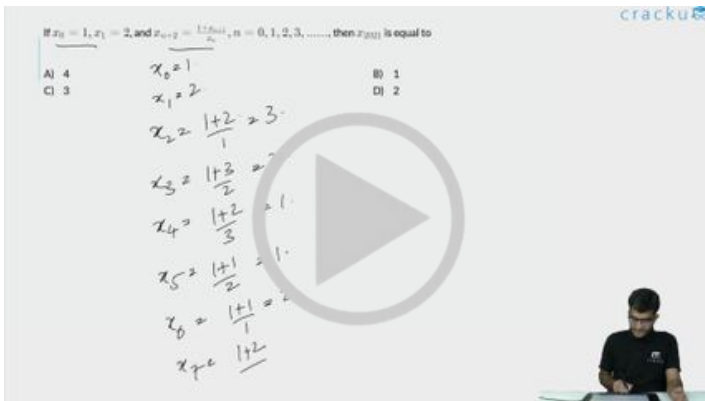


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Question 67

If $x_0 = 1, x_1 = 2$, and $x_{n+2} = \frac{1+x_{n+1}}{x_n}, n = 0, 1, 2, 3, \dots$, then x_{2021} is equal to

- A 4
- B 1
- C 3
- D 2



VIDEO SOLUTION

Question 68

If $a_1, a_2, \dots$ are in A.P., then, $\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_n}+\sqrt{a_{n+1}}}$ is equal to

- A $\frac{n}{\sqrt{a_1}+\sqrt{a_{n+1}}}$
- B $\frac{n-1}{\sqrt{a_1}+\sqrt{a_{n-1}}}$
- C $\frac{n-1}{\sqrt{a_1}+\sqrt{a_n}}$
- D $\frac{n}{\sqrt{a_1}-\sqrt{a_{n+1}}}$

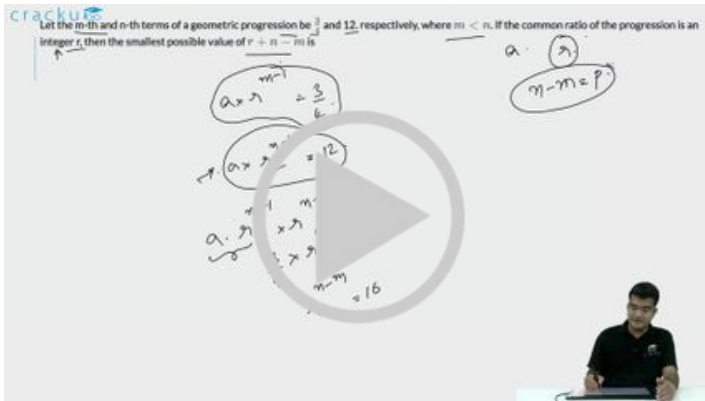


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Question 69

Let the m -th and n -th terms of a geometric progression be $\frac{3}{4}$ and 12, respectively, where $m < n$. If the common ratio of the progression is an integer r , then the smallest possible value of $r + n - m$ is

- A 6
- B 2
- C -4
- D -2



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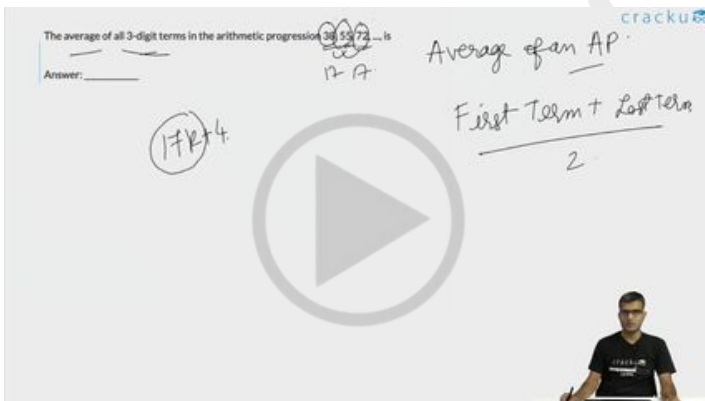
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Question 70

The average of all 3-digit terms in the arithmetic progression 38, 55, 72, ..., is



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Question 71

The value of $\frac{1}{1-x} + \frac{1}{1+x} + \frac{2}{1+x^2} + \frac{4}{1+x^4}$

- A $\frac{8}{1-x^8}$
- B $\frac{4x}{1+x^2}$

C $\frac{4}{1-x^6}$

D $\frac{4}{1+x^4}$

[illegible]

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Question 72

For any natural number n let a_n be the largest integer not exceeding $\sqrt{n}$. Then the value of $a_1 + a_2 + \dots + a_{50}$ is

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For any natural number n , let a_n be the largest integer not exceeding $\sqrt{n}$. Then the value of $a_1 + a_2 + \dots + a_{49}$ is

Answer: _____

$a_1 = 1$
 $a_2 = 1$
 $a_3 = 1$
 $a_4 = 2$
 $a_5 = 2$
 $a_6 = 2$
 $a_7 = 2$
 $a_8 = 2$
 $a_9 = 3$
 $a_{10} = 3$
 $a_{11} = 3$
 $a_{12} = 3$
 $a_{13} = 3$
 $a_{14} = 3$
 $a_{15} = 3$
 $a_{16} = 4$
 $a_{17} = 4$
 $a_{18} = 4$
 $a_{19} = 4$
 $a_{20} = 4$
 $a_{21} = 4$
 $a_{22} = 4$
 $a_{23} = 4$
 $a_{24} = 4$
 $a_{25} = 5$
 $a_{26} = 5$
 $a_{27} = 5$
 $a_{28} = 5$
 $a_{29} = 5$
 $a_{30} = 5$
 $a_{31} = 5$
 $a_{32} = 5$
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 $a_{37} = 6$
 $a_{38} = 6$
 $a_{39} = 6$
 $a_{40} = 6$
 $a_{41} = 6$
 $a_{42} = 6$
 $a_{43} = 6$
 $a_{44} = 6$
 $a_{45} = 6$
 $a_{46} = 6$
 $a_{47} = 6$
 $a_{48} = 6$
 $a_{49} = 7$

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Answers

1.A	2.14	3.D	4.C	5.C	6.B	7.B	8.D
9.A	10.D	11.D	12.A	13.D	14.A	15.A	16.B
17.80707	18.B	19.B	20.B	21.19	22.967	23.105	24.C
25.C	26.C	27.D	28.A	29.D	30.D	31.D	32.A
33.D	34.A	35.C	36.C	37.A	38.D	39.D	40.A
41.A	42.6144	43.A	44.B	45.C	46.A	47.C	48.C
49.A	50.4851	51.A	52.C	53.A	54.C	55.C	56.24
57.51	58.A	59.A	60.C	61.C	62.D	63.C	64.B
65.A	66.A	67.D	68.A	69.D	70.548	71.A	72.217

Explanations

1. A

Given, the number of particles on day 1 = 100

On day 2, one out of every 2 articles produces another particle.

The number of particles on day 2 will be $\frac{100}{2}$, i.e. 50 particles

On day 3, one out of every 3 articles produces another particle.

The number of particles on day 3 will be $\frac{100+50}{3}$, i.e. 50 particles

On day 4, one out of every 4 articles produces another particle.

The number of particles on day 4 will be $\frac{100+50+50}{4}$, i.e. 50 particles

Series will be 100, 50, 50, 50,....

It is given,

$$100 + (m-1)50 = 1000$$

$$m = 19$$

The answer is option A.

 VIDEO SOLUTION

2. 14

$$a_1 + a_2 + \dots + a_N = 300N$$

$$6a_1 + a_2 + \dots + a_N = 400N$$

$$5a_1 = 100N$$

$$a_1 = 20N$$

As the given sequence of numbers is non-decreasing sequence, N can take values from 2 to 15.

N is not equal to 1, if N = 1, then average of N numbers is 300 wouldn't satisfy.

Therefore, N can take values from 2 to 15, i.e. 14 values.

 VIDEO SOLUTION

3. D

The population of town at the beginning of 1st year = p

The population of town at the beginning of 2nd year = 3+2p

The population of town at the beginning of 3rd year = $2(3+2p)+3 = 2*2p+2*3+3 = 4p+3(1+2)$

The population of town at the beginning of 4th year = $2(2*2p+2*3+3)+3 = 8p+3(1+2+4)$

Similarly population at the beginning of the nth year = $2^{n-1}p+3(2^{n-1} - 1) = 2^{n-1}(p + 3)-3$

The population in the beginning of 2019 is 1000, then the population in the beginning of 2034 will be $(2^{2034-2019})(1000 + 3)-3 = 2^{15}(1003)-3$

[▶ VIDEO SOLUTION](#)

4. C

According to given conditions the terms are 81.33, -19, -100.33, -81.33, 19, 100.33, 81.33, -19,... Hence the series repeats after every 6 terms. Also summation of these 6 terms is 0. Hence summation is 60002 terms will we sum of first 2 terms which is 62.33.

[▶ VIDEO SOLUTION](#)

5. C

Assume the first series as $a, b, a/2, c, a+20$

and second series as x_1, x_2, x_3, x_4

$x_1=b-a, x_2= a/2-b, x_3=c-a/2$, and $x_4=a+20-c$

$x_2-x_1=30 \Rightarrow 3a-4b=60$

and $x_4-x_3=30 \Rightarrow 3a-4c=20$

and $x_4-x_2=60 \Rightarrow a-2c+2b=80$

Solving we get, $a=100, b=60$, and $c=70$

$S_1 = 100, 60, 50, 70, 120$

$S_2 = -40, -10, 20, 50$

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6. B

Assume the first series as $a, b, a/2, c, a+20$

and second series as x_1, x_2, x_3, x_4

$x_1=b-a, x_2= a/2-b, x_3=c-a/2$, and $x_4=a+20-c$

$x_2-x_1=30 \Rightarrow 3a-4b=60$

and $x_4-x_3=30 \Rightarrow 3a-4c=20$

and $x_4-x_2=60 \Rightarrow a-2c+2b=80$

Solving we get, $a=100, b=60$, and $c=70$

$S_1 = 100, 60, 50, 70, 120$

$S_2 = -40, -10, 20, 50$

Difference = 20

[▶ VIEW SOLUTION](#)

7. B

Let the first term of both series be a_1 , and b_1 , respectively, and the common difference be d_1 , and d_2 , respectively.

It is given that $a_5 = b_9$, which implies $a_1 + 4d_1 = b_1 + 8d_2$

$\Rightarrow a_1 - b_1 = 8d_2 - 4d_1 \dots \text{Eq(1)}$

Similarly, it is known that $a_{19}=b_{19}$, which implies $a_1 + 18d_1 = b_1 + 18d_2$

$$\Rightarrow a_1 - b_1 = 18d_2 - 18d_1 \dots \text{Eq(2)}$$

Equating (1) and (2), we get:

$$\Rightarrow 18d_2 - 18d_1 = 8d_2 - 4d_1$$

$$\Rightarrow 10d_2 = 14d_1$$

$$\Rightarrow 5d_2 = 7d_1$$

Since, d_1, d_2 are the prime numbers, which implies $d_1 = 5, d_2 = 7$.

It is also known that $b_2 = 0$, which implies $b_1 + d_2 = 0 \Rightarrow b_1 = -d_2 = -7$

Putting the value of b_1, d_1 , and, d_2 in Eq(1), we get:

$$a_1 = 8d_2 - 4d_1 + b_1 = 56 - 20 - 7 = 29$$

$$\text{Hence, } a_{11} = a_1 + 10d_1 = 29 + 10 \cdot 5 = 29 + 50 = 79$$

The correct option is B

 VIDEO SOLUTION

8. D

Now as per the given series :

$$\text{we get } x_1 = 1 + 2 = 3$$

$$\text{Now } x_1 - x_2 = 8$$

$$\text{so } x_2 = -5$$

$$\text{Now } x_1 - x_2 + x_3 = 15$$

$$\text{so } x_3 = 7$$

$$\text{so we get } x_n = (-1)^{n+1} (2n + 1)$$

$$\text{so } x_{49} = 99 \text{ and } x_{50} = -101$$

$$\text{Therefore } x_{49} + x_{50} = -2$$

 VIDEO SOLUTION

9. A

$$x_1 = -1$$

$$x_1 = x_2 + 2 \Rightarrow x_2 = x_1 - 2 = -3$$

Similarly,

$$x_3 = x_1 - 5 = -6$$

$$x_4 = -10$$

.

.

The series is -1, -3, -6, -10, -15.....

When the differences are in AP, then the n th term is $-\frac{n(n+1)}{2}$

$$x_{100} = -\frac{100(100+1)}{2} = -5050$$

 VIDEO SOLUTION

10. D

If the child adds all the numbers from 1 to 34, the sum of the numbers would be $1+2+3+\dots+34 = \frac{34 \times 35}{2} = 595$. Since the child got the sum as 575, he would have missed the number 20.

[▶ VIDEO SOLUTION](#)

CAT Previous Papers PDF

11. **D**

For the least element L in S_1 to be greater than the greatest element or equal to G in S_2 , the number that is added to L cannot be less than $G - L$.

[▶ VIEW SOLUTION](#)

12. **A**

We know that a_{24} is less than a_{25} .

So, even if a_{25} replaces a_{24} , the ascending order still exists in S_1 .

But, a_{25} is less than a_{26} . Hence, the descending order does not exist in S_2 anymore.

[▶ VIEW SOLUTION](#)

13. **D**

We will give an example to disprove each of the three options A, B and C and hence, the correct answer will be option D.

Initially, if the least integer in S_1 is -20 and the greatest integer in S_2 is 50, then after the doing the operations mentioned in the question, the greatest integer in S_1 is not greater than 50. Hence option A is false.

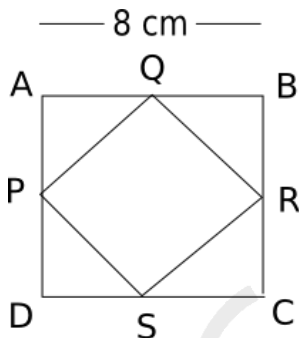
G is in S_2 as per the given information $\Rightarrow$ Option B is false.

If S_1 contains numbers from 1 to 24 and S_2 contains numbers from 25 to 50. Then $G = 50$ and $L = 1$. If all the numbers of S_1 change in sign, G will remain 50 and will be in S_2 while L will be -24 and will be in S_1 .

Hence, none of the statements is true always.

[▶ VIEW SOLUTION](#)

14. **A**



Side of first square = 8cm.

Side of second square made by joining mid-points of first square = $\frac{8}{\sqrt{2}}$

Similarly side of third square = $\frac{8}{\sqrt{2} \times \sqrt{2}}$ and so on.

Now summation of areas will be $8^2 + \left(\frac{8}{\sqrt{2}}\right)^2 + \left(\frac{8}{\sqrt{2} \times \sqrt{2}}\right)^2 + \dots$

$$\text{or } 8^2(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \dots)$$

$$\text{or } 64 \times (\frac{1}{1-\frac{1}{2}}) \text{ (As we know sum of an infinite G.P. is } \frac{a}{1-r} \text{ where a is first term and r is common ratio)}$$

or 128 sq. cm.

[VIEW SOLUTION](#)

15. A

$\frac{-p}{1-p}$ can be compared with sum of an infinite G.P. series i.e. $\frac{a}{1-r}$ (where a is first term and r is common ratio)

Hence here $a = (-p)$

and $r = p$

So k^{th} term will be $= (-p)(p)^{(k-1)}$

[VIEW SOLUTION](#)



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16. B

$a_1 = 3$ and $a_2 = 7$. Hence, the common difference of the AP is 4. If we assume, $k = 3n$

We have been given that the sum up to $3n$ terms of this AP is 1830. Hence, $1830 = \frac{k}{2}[2 * 3 + (k - 1) * 4]$

$$\Rightarrow 1830 * 2 = k(6 + 4k - 4)$$

$$\Rightarrow 3660 = 2k + 4k^2$$

$$\Rightarrow 2k^2 + k - 1830 = 0$$

$$\Rightarrow (k - 30)(2k + 61) = 0$$

$$\Rightarrow k = 30 \text{ or } k = -61/2$$

Since k is the number of terms so k cannot be negative. Hence, must be 30

$$\text{So, } 3n = 30$$

$$n = 10$$

$$\text{Sum of the first '10' terms of the given AP} = 5 * (6 + 9 * 4) = 42 * 5 = 210$$

$$m(a_1 + a_2 + \dots + a_n) > 1830$$

$$\Rightarrow 210m > 1830$$

$$\Rightarrow m > 8.71$$

Hence, smallest integral value of 'm' is 9.

[VIDEO SOLUTION](#)

17. 80707

$$S = 7 \times 11 + 11 \times 15 + 15 \times 19 + \dots + 95 \times 99$$

$$N^{\text{th}} \text{ term of the series can be written as } T_n = (4n + 3) * (4n + 7)$$

$$\text{Last term, } (4n+3) = 95 \text{ i.e. } n = 23$$

$$\sum_{n=1}^{n=23} (4n + 3) * (4n + 7)$$

$$\Rightarrow \sum_{n=1}^{n=23} 16n^2 + 40n + 21$$

$$\Rightarrow 16 * \frac{23 * 24 * 47}{6} + 40 * \frac{23 * 24}{2} + 21 * 23$$

$$\Rightarrow 80707$$

VIDEO SOLUTION

18. B

$$a_1 - a_2 + a_3 - a_4 + \dots + (-1)^{n-1} a_n = n$$

It is clear from the above equation that when n is odd, the co-efficient of a is positive otherwise negative.

$$a_1 - a_2 = 2$$

$$a_1 = a_2 + 2$$

$$a_1 - a_2 + a_3 = 3$$

On substituting the value of a_1 in the above equation, we get

$$a_3 = 1$$

$$a_1 - a_2 + a_3 - a_4 = 4$$

On substituting the values of a_1, a_3 in the above equation, we get

$$a_4 = -1$$

$$a_1 - a_2 + a_3 - a_4 + a_5 = 5$$

On substituting the values of a_1, a_3, a_4 in the above equation, we get

$$a_5 = 1$$

So we can conclude that $a_3, a_5, a_7 \dots a_{n+1} = 1$ and $a_2, a_4, a_6 \dots a_{2n} = -1$

Now we have to find the value of $a_{51} + a_{52} + \dots + a_{1023}$

$$\text{Number of terms} = 1023 - 51 + 1$$

$$n = 973$$

There will be 486 even and 487 odd terms, so the value of $a_{51} + a_{52} + \dots + a_{1023} = 486 \times -1 + 487 \times 1 = 1$

VIDEO SOLUTION

19. B

A: 15, 19, 23, 27, ..., 415

B: 14, 19, 24, 29, ..., 464

Here the first common term = 19

Common difference = LCM of 5, 4 = 20

$$19 + (n-1)20 \leq 415$$

$$(n-1)20 \leq 396$$

$$(n-1) \leq 19.8$$

$$n = 20$$

VIDEO SOLUTION

20. B

Given that $\frac{1}{\sqrt{y} + \sqrt{z}}$ is the arithmetic mean of $\frac{1}{\sqrt{x} + \sqrt{z}}$ and $\frac{1}{\sqrt{x} + \sqrt{y}}$

$$\Rightarrow \frac{2}{\sqrt{y} + \sqrt{z}} = \frac{1}{\sqrt{x} + \sqrt{z}} + \frac{1}{\sqrt{x} + \sqrt{y}}$$

$$\begin{aligned} \Rightarrow 2(\sqrt{x} + \sqrt{z})(\sqrt{x} + \sqrt{y}) &= (\sqrt{y} + \sqrt{z})(\sqrt{x} + \sqrt{y} + \sqrt{x} + \sqrt{z}) \\ \Rightarrow 2(x + \sqrt{xy} + \sqrt{xz} + \sqrt{yz}) &= 2\sqrt{xy} + y + \sqrt{yz} + 2\sqrt{xz} + \sqrt{yz} + z \\ \Rightarrow 2x &= y + z \\ \Rightarrow y, x, z &\text{ are in A.P. as } x \text{ is the arithmetic mean of } y \text{ and } z. \end{aligned}$$

 VIDEO SOLUTION

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21. 19

Given on day-1, there are 2 organisms.

On day-2, there are $2 \times 2 + 3 = 7$ and on day-3, there are $2 \times 7 + 3 = 17$.

Let us try to form a pattern:

$$2 = 2 + 0 \quad (n = 1)$$

$$7 = 4 + 3 \quad (n = 2)$$

$$17 = 8 + 9 \quad [8 + 3 \times 3] \quad (n = 3)$$

$$37 = 16 + 21 \quad [16 + 3 \times 7] \quad n = 4$$

$$T(n) = 2^n + 3(2^{n-1} - 1)$$

We know that $2^{20} = 2^{10} \times 2^{10} = 1024 \times 1024$, which is more than 1 million.

Let us check for $n = 19$

$$2^{19} + 3(2^{18} - 1) = 2^{19} + 3 \cdot 2^{18} - 3 = 2 \cdot 2^{19} + 2^{18} - 3 = 2^{20} + 2^{18} - 3, \text{ which is more than 1 million.}$$

Let us check for $n = 18$

$$\Rightarrow 2^{18} + 3(2^{17} - 1) = 2^{18} + 3 \cdot 2^{17} - 3 = 2 \cdot 2^{18} + 2^{17} - 3 = 2^{19} + 2^{17} - 3 \text{ which is not more than a million.}$$

$$\Rightarrow n = 19.$$

 VIDEO SOLUTION

22. 967

It is given that $a_n = 13 + 6(n - 1)$, which can be written as $a_n = 13 + 6n - 6 = 7 + 6n$

Similarly, $b_n = 15 + 7(n - 1)$, which can be written as $b_n = 15 + 7n - 7 = 8 + 7n$

The common differences are 6, and 7, respectively, The common difference of terms that exists in both series is l.c.m (6, 7) = 42

The first common term of the first two series is 43 (by inspection)

Hence, we need to find the m th term, which is less than 1000, and the largest three-digit integer, and exists in both series.

$$t_m = a + (m - 1)d < 1000$$

$$\Rightarrow 43 + (m - 1)42 < 1000$$

$$\Rightarrow (m - 1)42 < 957$$

$$\Rightarrow m - 1 < 22.8 \Rightarrow m < 23.8 \Rightarrow m = 23$$

$$\text{Hence, the 23rd term is } 43 + 22 \times 42 = 967$$

VIDEO SOLUTION

23.105

Given that the arithmetic mean of x, y and z is 80.

$$\Rightarrow \frac{x + y + z}{3} = 80$$

$$\Rightarrow x + y + z = 240 \dots (1)$$

$$\text{Also, } \frac{x + y + z + v + u}{5} = 75$$

$$\Rightarrow \frac{x + y + z + v + u}{5} = 75$$

$$\Rightarrow x + y + z + v + u = 375$$

Substituting values from equation (1),

$$\Rightarrow v + u = 135$$

It is given that $u = (x+y)/2$ and $v = (y+z)/2$.

$$\Rightarrow (x + y)/2 + (y + z)/2 = 135$$

$$\Rightarrow x + 2y + z = 270$$

$$\Rightarrow y = 30 \text{ (Since } x + y + z = 240)$$

Therefore, we can say that $x + z = 240 - y = 210$. We are also given that $x \geq z$,

Hence, $x_{\min} = 210/2 = 105$.

VIDEO SOLUTION

24. C

$$x = \sqrt{4 + \sqrt{4 - \sqrt{4 + \sqrt{4 - \dots \text{to infinity}}}}}$$

$$\Rightarrow x = \sqrt{4 + \sqrt{4 - x}}$$

$$\Rightarrow x^2 = 4 + \sqrt{4 - x}$$

$$\Rightarrow x^4 + 16 - 8x^2 = 4 - x$$

$$\Rightarrow x^4 - 8x^2 + x + 12 = 0$$

On substituting options, we can see that option C satisfies the equation.

VIEW SOLUTION

25. C

Let $x = a, y = ar$ and $z = ar^2$

It is given that, $5x, 16y$ and $12z$ are in AP.

so, $5x + 12z = 32y$

On replacing the values of x, y and z , we get

$$5a + 12ar^2 = 32ar$$

$$\text{or, } 12r^2 - 32r + 5 = 0$$

On solving, $r = \frac{5}{2}$ or $\frac{1}{6}$

For $r = \frac{1}{6}$, $x < y < z$ is not satisfied.

So, $r = \frac{5}{2}$

Hence, option C is the correct answer.

 VIDEO SOLUTION



26. **C**

The terms of the first sequence are of the form $4p + 13$

The terms of the second sequence are of the form $5q + 11$

If a term is common to both the sequences, it is of the form $4p+13$ and $5q+11$

or $4p = 5q - 2$. LHS = $4p$ is always even, so, q is also even.

or $2p = 5r - 1$ where $q = 2r$.

Notice that LHS is again even, hence r should be odd. Let $r = 2m+1$ for some m .

Hence, $p = 5m + 2$.

So, the number = $4p+13 = 20m + 21$.

Hence, all numbers of the form $20m + 21$ will be the common terms. i.e $21, 41, 61, \dots, 401 = 20$.

 VIDEO SOLUTION

27. **D**

1, 2, 3, 4, ..., n such that the sum is greater than 288

If $n = 24$, $n(n+1)/2 = 12 \cdot 25 = 300$

So, $n = 24$, i.e. the 24th letter in the alphabet is the letter at position 288 in the series

So, answer = x

 VIDEO SOLUTION

28. **A**

Let there only be 2 questions.

Thus there are $2^{2-1} = 2$ students who have done 1 or more questions wrongly, $2^{2-2} = 1$ students who have done all 2 questions wrongly.

Thus total number of wrong answers = $2 + 1 = 3 = 2^n - 1$.

Now let there be 3 questions. Then $j = 1, 2, 3$

Number of students answering 1 or more questions incorrectly = 4

Number of students answering 2 or more questions incorrectly = 2

Number of students answering 3 or more questions incorrectly = 1

Total number of incorrect answers = $1(3) + (2-1) \cdot 2 + (4-2) \cdot 1 = 7 = 2^3 - 1$

According to the question, the total number of wrong answers = $4095 = 2^{12} - 1$.

Hence Option A.

 VIEW SOLUTION

29. **D**

Let x be in the front row.

So no. of children in next rows will be $x-3, x-6, x-9, x-12, x-15, x-18, x-21, \dots$

Suppose there are 6 rows, then the sum is equal to $x + x-3 + x-6 + x-9 + x-12 + x-15 = 6x - 45$

This sum is equal to 630.

$$\Rightarrow 6x - 45 = 630 \Rightarrow 6x = 675$$

Here, x is not an integer.

Hence, there cannot be 6 rows.

 VIDEO SOLUTION

30. D

$$a_2 = 2 * 1 + 5$$

$$a_3 = 2 * (2 + 5) + 5 = 2^2 + 5 * 2 + 5$$

$$a_4 = 2^3 + 5 * 2^2 + 5 * 2 + 5$$

...

$$\begin{aligned} a_{100} &= 2^{99} + 5 * (2^{98} + 2^{97} + \dots + 1) \\ &= 2^{99} + 5 * 1 * (2^{99} - 1) / (2 - 1) = 2^{99} + 5 * 2^{99} - 5 = 6 * 2^{99} - 5 \end{aligned}$$

 VIDEO SOLUTION

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31. D

Let $x = 1$, so, $X_0 = 1$

$$X_1 = -1$$

$$X_2 = -1$$

$$X_3 = 1$$

$$X_4 = 1$$

$$X_5 = -1$$

$$X_6 = -1$$

So, X_n need not be positive when n is even, X_n need not be positive when n is odd, X_n need not be negative when n is even. So, none of the first three options are correct.

 VIEW SOLUTION

32. A

$$\text{Let } S = 2 + 5x + 9x^2 + \dots$$

$$S * x = 2x + 5x^2 + 9x^3 + \dots$$

$$S(1 - x) = 2 + 3x + 4x^2 + \dots$$

$$S(1 - x) * x = 2x + 3x^2 + 4x^3 + \dots$$

$$S(1 - x)(1 - x) = 2 + x + x^2 + x^3 + \dots = 2 + x/(1 - x)$$

$$\text{So, } S = [2(1 - x) + x]/(1 - x)^3 \Rightarrow S = (2 - x)/(1 - x)^3$$

 VIEW SOLUTION

33. D

$$\text{Given } x_{n+1} = x_n + n - 1 \text{ and } x_1 = -1.$$

Considering

$$x_1 = -1. \quad (1)$$

$$x_2 = x_1 + 1 - 1 = x_1 + 0 \quad (2)$$

$$x_3 = x_2 + 2 - 1 = x_2 + 1 \quad (3)$$

$$x_4 = x_3 + 3 - 1 = x_3 + 2 \quad (4)$$

$$x_{100} = x_{99} + 98 \quad (100)$$

Adding the LHS and RHS for the hundred equations we have:

$$(x_1 + x_2 + \dots + x_{100}) = (-1 + 0 + \dots + 98) + (x_1 + x_2 + \dots + x_{99})$$

Subtracting this we have :

$$(x_1 + \dots + x_{100}) - (x_1 + \dots + x_{99}) = \frac{(98 \cdot 99)}{2} - 1.$$

$$x_{100} = 4851 - 1 = 4850$$

Alternatively

$$x_1 = -1$$

$$x_2 = x_1 + 1 - 1 = x_1 = -1$$

$$x_3 = x_2 + 2 - 1 = x_2 + 1 = -1 + 1 = 0$$

$$x_4 = x_3 + 3 - 1 = x_3 + 2 = 0 + 2 = 2$$

$$x_5 = x_4 + 4 - 1 = x_4 + 3 = 2 + 3 = 5$$

.....

If we observe the series, it is a series that has a difference between the consecutive terms in an AP.

Such series are represented as $t(n) = a + bn + cn^2$

We need to find $t(100)$.

$$t(1) = -1$$

$$a + b + c = -1$$

$$t(2) = -1$$

$$a + 2b + 4c = -1$$

$$t(3) = 0$$

$$a + 3b + 9c = 0$$

Solving we get,

$$b + 3c = 0$$

$$b + 5c = 1$$

$$c = 0.5$$

$$b = -1.5$$

$$a = 0$$

Now,

$$t(100) = (-1.5) 100 + (0.5) 100^2 = -150 + 5000 = 4850$$

 VIDEO SOLUTION

34. A

$$a_{100} = \frac{1}{(3 \times 100 - 1) \times (3 \times 100 + 2)} = \frac{1}{299 \times 302}$$

$$\frac{1}{2 \times 5} = \frac{1}{3} \times \left(\frac{1}{2} - \frac{1}{5} \right)$$

$$\frac{1}{5 \times 8} = \frac{1}{3} \times \left(\frac{1}{5} - \frac{1}{8} \right)$$

$$\frac{1}{8 \times 11} = \frac{1}{3} \times \left(\frac{1}{8} - \frac{1}{11} \right)$$

....

$$\frac{1}{299 \times 302} = \frac{1}{3} \times \left(\frac{1}{299} - \frac{1}{302} \right)$$

$$\begin{aligned} \text{Hence } a_1 + a_2 + a_3 + \dots + a_{100} &= \frac{1}{3} \times \left(\frac{1}{2} - \frac{1}{5} \right) + \frac{1}{3} \times \left(\frac{1}{5} - \frac{1}{8} \right) + \frac{1}{3} \times \left(\frac{1}{8} - \frac{1}{11} \right) + \dots + \frac{1}{3} \times \left(\frac{1}{299} - \frac{1}{302} \right) \\ &= \frac{1}{3} \times \left(\frac{1}{2} - \frac{1}{302} \right) \\ &= \frac{25}{151} \end{aligned}$$

 VIDEO SOLUTION

35. C

Let the common ratio of the G.P. be r .

Hence we have $a_n = 3(a_{n+1} + a_{n+2} + \dots)$

The sum up to infinity of GP is given by $\frac{a}{1-r}$ where a here is a_{n+1}

$$\Rightarrow a_n = 3 \left(\frac{a_{n+1}}{1-r} \right)$$

$$\Rightarrow a_n = 3 \left(\frac{a_n \times r}{1-r} \right)$$

$$\Rightarrow r = \frac{1}{4}$$

Now, $a_1 + a_2 + a_2 \dots = 32$

$$\Rightarrow \frac{a_1}{1-r} = 32$$

$$\Rightarrow \frac{a_1}{3/4} = 32$$

$$\Rightarrow a_1 = 24$$

$$a_5 = a_1 \times r^4$$

$$a_5 = 24 \times (1/4)^4 = \frac{3}{32}$$

 VIDEO SOLUTION



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36. C

It is given,

$$S_n = 2n^2 + n$$

$$S_{n-1} = 2(n-1)^2 + (n-1)$$

$$S_{n-1} = 2n^2 - 3n + 1$$

$$T_n = S_n - S_{n-1} = 2n^2 + n - 2n^2 + 3n - 1 = 4n - 1$$

$$T_n = 4n - 1$$

The terms are 3, 7, 11, 15, 19, 23, 27,.....

27 is the first term in the series divisible by 9.

27 is the 7th term.

Therefore, the least possible value of n is 7.

The answer is option C.

 VIDEO SOLUTION

37. A

Sum of n terms in an A.P. = $\frac{n}{2} (2a + (n-1)d)$

$$A_n = \frac{n}{2} (6 + (n-1)4) = n(2n+1)$$

$$\Sigma A_n = \Sigma n(2n+1) = 2\Sigma n^2 + \Sigma n = \frac{2n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$$

Substituting $n = 25$, we get

$$\frac{1}{25} \sum_{n=1}^{25} A_n = \frac{1}{25} \left(\frac{2(25)(25+1)(50+1)}{6} + \frac{25(25+1)}{2} \right)$$

$$\frac{1}{25} \sum_{n=1}^{25} A_n = 26(17) + 13 = 455$$

The answer is option A.

 VIDEO SOLUTION

38. D

It is given that $\frac{1}{2}$, $\frac{\log_3(2^x-9)}{\log_3 4}$, and $\frac{\log_5(2^x+\frac{17}{2})}{\log_5 4}$ are in an arithmetic progression.

$\frac{\log_3(2^x-9)}{\log_3 4}$ can be written as $\log_4(2^x-9)$, and $\frac{\log_5(2^x+\frac{17}{2})}{\log_5 4}$ can be written as $\log_4(2^x+\frac{17}{2})$

$$\text{Hence, } 2\log_4(2^x-9) = \frac{1}{2} + \log_4(2^x+\frac{17}{2})$$

$\frac{1}{2}$ can be written as $\log_4 2$.

Therefore,

$$\Rightarrow 2\log_4(2^x-9) = \log_4 2 + \log_4(2^x+\frac{17}{2})$$

$$\Rightarrow \log_4(2^x-9)^2 = \log_4 2(2^x+\frac{17}{2})$$

$$\Rightarrow (2^x-9)^2 = 2(2^x+\frac{17}{2})$$

$$\Rightarrow 2^{2x} - 18 \cdot 2^x + 81 = 2 \cdot 2^x + 17$$

$$\Rightarrow 2^{2x} - 20 \cdot 2^x + 64 = 0$$

$$\Rightarrow 2^{2x} - 16 \cdot 2^x - 4 \cdot 2^x + 64 = 0$$

$$\Rightarrow 2^x(2^x-16) - 4(2^x-16) = 0$$

$$\Rightarrow (2^x-4)(2^x-16) = 0$$

The values of 2^x can't be 4 (log will be undefined), which implies The value of 2^x is 16.

Therefore, the common difference is $\log_4(2^x-9) - \log_4 2$

$$\Rightarrow \log_4 7 - \log_4 2 = \log_4\left(\frac{7}{2}\right)$$

The correct option is D

 VIDEO SOLUTION

39. D

The given sequence can be written as:

$$1\left(1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots\right) + \frac{1}{3}\left(\frac{1}{4} + \frac{1}{16} + \dots\right) + \frac{1}{9}\left(\frac{1}{16} + \frac{1}{64} + \dots\right) + \dots$$

We know that the sum of an infinite G.P. is $\frac{a}{1-r}$, where a is the first term and r is the common ratio.

$$\Rightarrow \text{The first term} = \frac{1}{1-\frac{1}{4}} = \frac{4}{3}$$

$$\Rightarrow \text{The second term} = \frac{1}{3} \left(\frac{\left(\frac{1}{4}\right)}{1-\left(\frac{1}{4}\right)} \right) = \frac{1}{9}$$

$$\Rightarrow \text{The third term} = \frac{1}{9} \left(\frac{\left(\frac{1}{16}\right)}{1-\left(\frac{1}{4}\right)} \right) = \frac{1}{108}$$

Observing these three terms, we see that they are in G.P. with a common ratio of $\frac{1}{12}$

$$\Rightarrow \text{Sum of this infinite G.P.} = \frac{\left(\frac{4}{3}\right)}{1 - \left(\frac{1}{12}\right)} = \frac{16}{11}$$

[▶ VIDEO SOLUTION](#)

40. **A**

The first series goes as follows:

46, 54, 62, 70, 78, 86, 94, 102,....

The second series goes as follows:

98, 102, 106, 110,...

The first common term is 102 (first term of the common terms) and the common difference between them will be $\text{hcf}(4,8) = 8$

$\Rightarrow$ The required sequence is 102, 110, 118,..... (last term should be less than 468 (100th term of second series))

$$\Rightarrow 102 + (n-1)(8) \leq 498$$

$$\Rightarrow n \text{ is less than or equal to } 50.5 \Rightarrow n = 50$$

Using the summation of A.P. formula:

$$\text{Required sum} = \frac{n}{2} (2a + (n-1)d) = \frac{50}{2} (2 \times 102 + 49 \times 8) = 14900$$

[▶ VIDEO SOLUTION](#)

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41. **A**

Suppose consecutive odd integers are: $(a-2)$, a , $(a+2)$

$$\text{Hence, } 3a-6 = 2(a+2) + 3 \Rightarrow a=13$$

$$a+2 = 15$$

[▶ VIEW SOLUTION](#)

42. **6144**

$$11\text{th term of series} = a_{11} = \text{Sum of 11 terms} - \text{Sum of 10 terms} = 3(2^{11+1} - 2) - 3(2^{10+1} - 2)$$

$$= 3(2^{12} - 2 - 2^{11} + 2) = 3(2^{11})(2 - 1) = 3 \cdot 2^{11} = 6144$$

[▶ VIDEO SOLUTION](#)

43. **A**

Using the arithmetic progression formula for the n th term, where

$$x_n = a + (n-1)d$$

Substituting the value for n and using that in the equation that is given,

$$2x_6 + 2x_9 = x_{11} + x_{13}, \text{ Then, } x_{100} \text{ equals}$$

$$\text{We get, } 2(a + 5d) + 2(a + 8d) = a + 10d + a + 12d$$

$$4a + 26d = 2a + 22d$$

$$2a = -4d$$

$$a = -2d$$

We are given, $x_5 = -4$

$$a + 4d = -4$$

Substituting the value for a in terms of d,

$$2d = -4$$

$$d = -2$$

$$a = 4$$

$$x_{100} = a + 99d$$

$$x_{100} = 4 - 198 = -194$$

 VIDEO SOLUTION

44. B

Opening the brackets, we get the series as: $\left(\frac{1}{5}\right)^2 - \left(\frac{1}{5} \times \frac{1}{7}\right) + \left(\frac{1}{5}\right)^4 - \left(\frac{1}{5} \times \frac{1}{7}\right)^2 + \left(\frac{1}{5}\right)^6 - \left(\frac{1}{5} \times \frac{1}{7}\right)^6 + \dots$

These are two infinite GPs when rearranged:

$$\left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^4 + \left(\frac{1}{5}\right)^6 + \dots - \left(\frac{1}{5} \times \frac{1}{7}\right) - \left(\frac{1}{5} \times \frac{1}{7}\right)^6 - \left(\frac{1}{5} \times \frac{1}{7}\right)^2 - \dots$$

The sum of the first series would be $\frac{\frac{1}{25}}{1 - \frac{1}{25}} = \frac{1}{24}$

The sum of the second series would be $\frac{\frac{1}{35}}{1 - \frac{1}{35}} = \frac{1}{34}$

The answer to the given series would then be $\frac{1}{24} - \frac{1}{34} = \frac{10}{816} = \frac{5}{408}$

Therefore, Option B is the correct answer.

 VIDEO SOLUTION

45. C

The sum of the 3rd and 15th terms is $a+2d+a+14d = 2a+16d$

The sum of the 6th, 11th and 13th terms is $a+5d+a+10d+a+12d = 3a+27d$

Since the two are equal, $2a+16d = 3a+27d \Rightarrow a+11d = 0$

So, the 12th term is 0

 VIDEO SOLUTION

Free CAT Study Material

46. A

Finding the terms in the sequence, we see that $t_3 = 0, t_4 = -\frac{1}{3}, t_5 = 0$

We would notice that all the odd terms are 0, and we are also asked the sum of only even terms, so we do not need to consider those

$$t_6 = -\frac{1}{5}$$

We see that the even terms are in an HP: $-1, -\frac{1}{3}, -\frac{1}{5}, -\frac{1}{7}, \dots$

The sum we are asked is the inverse of these terms, that is: $-1, -3, -5, -7$, up to 1012 terms

The sum of this AP would be $\frac{[-(2 \times 1) + (1012 - 1)(-2)]}{2} \times 1012$

Which is equal to $-1012 \times 1012 = -1024144$

Therefore, Option A is the correct answer.

 VIDEO SOLUTION

47. C

Assume the first series as $a, b, a/2, c, a+20$

and second series as x_1, x_2, x_3, x_4

$x_1 = b - a, x_2 = a/2 - b, x_3 = c - a/2$, and $x_4 = a + 20 - c$

$x_2 - x_1 = 30 \Rightarrow 3a - 4b = 60$

and $x_4 - x_3 = 30 \Rightarrow 3a - 4c = 20$

and $x_4 - x_2 = 60 \Rightarrow a - 2c + 2b = 80$

Solving we get, $a = 100, b = 60$, and $c = 70$

$S_1 = 100, 60, 50, 70, 120$

$S_2 = -40, -10, 20, 50$

Avg. = $(100 + 60 + 50 + 70 + 120) \div 5 = 80$

 VIEW SOLUTION

48. C

$(1/(2^2 - 1)) + (1/(4^2 - 1)) + (1/(6^2 - 1)) + \dots + (1/(20^2 - 1)) = 1/[(2+1)(2-1)] + 1/[(4+1)(4-1)] + \dots + 1/[(20+1)(20-1)]$

$= 1/(1 \cdot 3) + 1/(3 \cdot 5) + 1/(5 \cdot 7) + \dots + 1/(19 \cdot 21)$

$= 1/2 * (1/1 - 1/3 + 1/3 - 1/5 + 1/5 - 1/7 + \dots + 1/19 - 1/21)$

$= 1/2 * (1 - 1/21) = 10/21$

 VIDEO SOLUTION

49. A

The seventh term of an AP = $a + 6d$. Third term will be $a + 2d$ and second term will be $a + 16d$. We are given that

$(a + 6d)^2 = (a + 2d)(a + 16d)$

$\Rightarrow a^2 + 36d^2 + 12ad = a^2 + 18ad + 32d^2$

$\Rightarrow 4d^2 = 6ad$

$\Rightarrow d : a = 3 : 2$

We have been asked about $a:d$. Hence, it would be 2:3

 VIDEO SOLUTION

50. 4851

Let us first find the number of terms

$47 = 1 + (n-1)2$

$n = 24$

$$24 \cdot 2n + 1 + 3 + 5 + \dots + 47 = 5280$$

$$48n + 576 = 5280$$

$$48n = 4704$$

$$n = 98$$

$$\text{Sum of first 98 terms} = 98 \cdot 99 / 2$$

$$= 4851$$

 VIDEO SOLUTION

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51. A

The first number in each group: 1, 2, 5, 10, 17,

Their common difference is in Arithmetic Progression. Hence, the general term of the series can be expressed as a quadratic equation.

Let the quadratic equation of the general term be $ax^2 + bx + c$

$$1\text{st term} = a + b + c = 1$$

$$2\text{nd term} = 4a + 2b + c = 2$$

$$3\text{rd term} = 9a + 3b + c = 5$$

Solving the equations, we get $a=1$, $b=-2$, $c=2$.

$$\text{Hence, the first term in the 15th group will be } (15)^2 - 2(15) + 2 = 197$$

We can see that the number of terms in each group is 1, 3, 5, 7, and so on. These are of the form $2n-1$. Hence, the number of terms in 15th group will be 29. Hence, the last term in the 15th group will be $197 + 29 - 1 = 225$.

$$\text{Sum of terms in group 15} = \frac{29}{2} (197 + 225) = 6119$$

Alternatively,

The final term in each group is the square of the group number.

In the first group 1, second group 4,

$$\text{The final element of the 14th group is } (14)^2 = 196, \text{ similarly for the 15th group this is: } (15)^2 = 225$$

Each group contains all the consecutive elements in this range.

Hence the 15th group the elements are:

$$(197, 198, \dots, 225).$$

This is an Arithmetic Progression with a common difference of 1 and the number of element 29.

$$\text{Hence the sum is given by: } \frac{n}{2} \cdot (\text{first term} + \text{last term})$$

$$\frac{29}{2} \cdot (197 + 225)$$

$$6119.$$

 VIDEO SOLUTION

52. C

Let 'x' be the average of all 52 positive integers $a_1, a_2, \dots, a_{52}$.

$$a_1 + a_2 + a_3 + \dots + a_{52} = 52x \dots (1)$$

Therefore, average of $a_2, a_3, \dots, a_{52} = x+1$

$$a_2 + a_3 + a_4 + \dots + a_{52} = 51(x+1) \dots (2)$$

From equation (1) and (2), we can say that

$$a_1 + 51(x + 1) = 52x$$

$$a_1 = x - 51.$$

We have to find out the largest possible value of a_1 . a_1 will be maximum when 'x' is maximum.

$(x+1)$ is the average of terms $a_2, a_3, \dots, a_{52}$. We know that $a_2 < a_3 < \dots < a_{52}$ and $a_{52} = 100$.

Therefore, $(x+1)$ will be maximum when each term is maximum possible. If $a_{52} = 100$, then $a_{51} = 99, a_{50} = 98$ ends so on.

$$a_2 = 100 + (51-1)*(-1) = 50.$$

$$\text{Hence, } a_2 + a_3 + a_4 + \dots + a_{52} = 50 + 51 + \dots + 99 + 100 = 51(x+1)$$

$$\Rightarrow \frac{51 * (50 + 100)}{2} = 51(x + 1)$$

$$\Rightarrow x = 74$$

Therefore, the largest possible value of $a_1 = x - 51 = 74 - 51 = 23$.

 VIDEO SOLUTION

53. A

substituting 3,4...53 in the given function, we get

$$\begin{aligned} t_3 &= \frac{3}{5} \\ t_4 &= \frac{4}{6} \\ t_5 &= \frac{5}{7} \\ t_6 &= \frac{6}{8} \end{aligned}$$

Multiplying the values, we get $\frac{3}{5} * \frac{4}{6} * \frac{5}{7} * \dots * \frac{52}{54} * \frac{53}{55}$ which ultimately after cancellations give $\frac{3*4}{54*55} = \frac{2}{495}$

 VIDEO SOLUTION

54. C

Given x, y, z are three terms in an arithmetic progression.

Considering $x = a, y = a+d, z = a+2*d$.

Using the given equation $x*y*z = 5*(x+y+z)$

$$a*(a+d)*(a+2*d) = 5*(a+a+d+a+2*d)$$

$$= a*(a+d)*(a+2*d) = 5*(3*a+3*d) = 15*(a+d).$$

$$= a*(a+2*d) = 15.$$

Since all x, y, z are positive integers and $y-x > 2$. $a, a+d, a+2*d$ are integers.

The common difference is positive and greater than 2.

Among the different possibilities are : $(a=1, a+2d = 5), (a=3, a+2d = 5), (a=5, a+2d = 3), (a=15, a+2d = 1)$

Hence the only possible case satisfying the condition is :

$$a = 1, a+2*d = 15.$$

$$x = 1, z = 15.$$

$$z-x = 14.$$

 VIDEO SOLUTION

55. C

It is given that in a Fibonacci sequence, from the third term onwards, each term in the sequence is the sum of the previous two terms in that sequence.

Let x and y be the 1st and 2nd term respectively.

3rd term = $x+y$

4th term = $x+2y$

5th term = $2x+3y$

6th term = $3x+5y$

7th term = $5x+8y$

We know that difference of the squares of 6th and 7th terms is $517 = 47 \times 11$.

And $a^2 - b^2 = (a + b)(a - b)$.

Applying above formula we get $(8x+13y)(2x+3y) = 47 \times 11$.

So only possible way is $(8x+13y)=47$ and

$2x+3y=11$.

Solving we get $x=1$ and $y=3$.

Using the concept that every term is the sum of the previous two terms, as used in the beginning of the solution, we get 10th term as $21x+34y$, which gives 10th term as 123.

 VIDEO SOLUTION

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56.24

It is given that $t_1 + t_2 + \dots + t_n = 2n^2 + 9n + 13$, for every positive integer $n \geq 2$.

We can say that $t_1 + t_2 + \dots + t_k = 2k^2 + 9k + 13 \dots (1)$

Replacing k by $(k-1)$ we can say that

$$t_1 + t_2 + \dots + t_{k-1} = 2(k-1)^2 + 9(k-1) + 13 \dots (2)$$

On subtracting equation (2) from equation (1)

$$\Rightarrow t_k = 2k^2 + 9k + 13 - 2(k-1)^2 + 9(k-1) + 13$$

$$\Rightarrow 103 = 4k + 7$$

$$\Rightarrow k = 24$$

 VIDEO SOLUTION

57.51

Sum of the sequence of even numbers is $2a_3 + (2a_3 - 2) + (2a_3 - 4) + (2a_3 - 6) + (2a_3 - 8) = 450$

$$\Rightarrow 10a_3 - 20 = 450$$

$$\Rightarrow a_3 = 47$$

$$\text{Hence } a_5 = 47 + 4 = 51$$

 VIDEO SOLUTION

58.A

According to given condition we have ,

$$n+6b = nr^6 \text{ and } b=10.5n ,$$

$$63n + n = nr^6$$

$$r^6 = 64$$

$$r = 2$$

 VIDEO SOLUTION

59. **A**

January 1, 1950 to December 31, 1959 is a period of 10 years or 20 half years.
The person X after 1st year gets Rs. 300 in next year he gets Rs. 330 and so on.
So his earning is in AP with 10 $300+330+360+\dots$

Similarly earning of Y is in AP with 20 terms $200+215+230+245+\dots$

So, the total earnings of X equals $12 \times (300+330+\dots 10 \text{ terms}) = 52200$

The total earnings of Y equals $6 \times (200+215+230+\dots 20 \text{ terms}) = 41100$

So, the total earnings of the two equals $52200+41100 = 93300$

 VIDEO SOLUTION

60. **C**

For the given problem ,

$$\sum n(n+1)/2 = 8436 \text{ which is}$$

$$\sum n^2/2 + \sum n/2 = 8436 \text{ which is equal to}$$

$$n \times (n+1)(2n+1)/12 + n \times (n+1)/4 = 8436, \text{ solving we get } n=36.$$

Solving the equation might be lengthy. you can substitute the values in the options to arrive at the answer.

 VIDEO SOLUTION

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61. **C**

Let the numbers be $a_1, a_2, \dots, a_n$.

Since the numbers are positive,

$$AM \geq GM$$

$$\frac{a_1 + a_2 + a_3 + \dots + a_n}{n} \geq (a_1 \times a_2 \times \dots \times a_n)^{1/n}$$

$$a_1 + a_2 + a_3 + \dots + a_n \geq n$$

 VIDEO SOLUTION

62. **D**

According to given question S_{50} will have 50 terms

And its first term will be 50th number in the series $1, 2, 4, 7, \dots, T_{50}$

$$T_1 = 1$$

$$T_2 = 1 + 1$$

$$T_3 = 1 + 1 + 2$$

$$T_4 = 1 + 1 + 2 + 3$$

$$T_n = 1 + (1 + 2 + 3 + 4 + 5 + \dots + (n-1))$$

$$= 1 + \frac{n(n-1)}{2}$$

So $T_{50} = 1 + 1225 = 1226$

Hence $S_{50} = (1226, 1227, 1228, 1229, \dots)$

And summation will be $= \frac{50}{2} (2 \times 1226 + 49 \times 1) = 62525$

 VIDEO SOLUTION

63. C

Given series can be written as:

$$\sum_{n=1}^{100} \left(\frac{1}{n \times (n+1)} \right)$$

$$\text{or } \sum_{n=1}^{100} \left(\frac{(n+1)-n}{n \times (n+1)} \right)$$

$$\text{or } \sum_{n=1}^{100} \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

After putting values of n from 1 to 100, all terms will cancel out, only first and last terms will be there

i.e. $1 - \frac{1}{101}$

or $\frac{100}{101}$

 VIDEO SOLUTION

64. B

Assume the first series as $a, b, a/2, c, a+20$

and second series as x_1, x_2, x_3, x_4

$x_1 = b - a, x_2 = a/2 - b, x_3 = c - a/2, \text{ and } x_4 = a + 20 - c$

$x_2 - x_1 = 30 \Rightarrow 3a - 4b = 60$

and $x_4 - x_3 = 30 \Rightarrow 3a - 4c = 20$

and $x_4 - x_2 = 60 \Rightarrow a - 2c + 2b = 80$

Solving we get, $a = 100, b = 60, \text{ and } c = 70$

$S_1 = 100, 60, 50, 70, 120$

$S_2 = -40, -10, 20, 50$

Sum = 20

 VIEW SOLUTION

65. A

Assume the first series as $a, b, a/2, c, a+20$

and second series as x_1, x_2, x_3, x_4

$x_1 = b - a, x_2 = a/2 - b, x_3 = c - a/2, \text{ and } x_4 = a + 20 - c$

$x_2 - x_1 = 30 \Rightarrow 3a - 4b = 60$

and $x_4 - x_3 = 30 \Rightarrow 3a - 4c = 20$

and $x_4 - x_2 = 60 \Rightarrow a - 2c + 2b = 80$

Solving we get, $a = 100, b = 60, \text{ and } c = 70$

$S_1 = 100, 60, 50, 70, 120$

$S_2 = -40, -10, 20, 50$

 VIEW SOLUTION

CAT Percentile Predictor

66. A

Sum of the first 11 terms = $11/2 (2a+10d)$

Sum of the first 19 terms = $19/2 (2a+18d)$

$$\Rightarrow 22a+110d = 38a+342d \Rightarrow 16a = -232d$$

$$\Rightarrow 2a = -232/8 d = -29d$$

Sum of the first 30 terms = $15(2a+29d) = 0$

 VIDEO SOLUTION

67. D

$$x_0 = 1$$

$$x_1 = 2$$

$$x_2 = \frac{(1+x_1)}{x_0} = \frac{(1+2)}{1} = 3$$

$$x_3 = \frac{(1+x_2)}{x_1} = \frac{(1+3)}{2} = 2$$

$$x_4 = \frac{(1+x_3)}{x_2} = \frac{(1+2)}{3} = 1$$

$$x_5 = \frac{(1+x_4)}{x_3} = \frac{(1+1)}{2} = 1$$

$$x_6 = \frac{(1+x_5)}{x_4} = \frac{(1+1)}{1} = 2$$

Hence, the series begins to repeat itself after every 5 terms. Terms whose number is of the form $5n$ are 1, $5n+1$ are 2... and so on, where $n=0,1,2,3,\dots$

2021 is of the form $5n+1$. Hence, its value will be 2.

 VIDEO SOLUTION

68. A

We have, $\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_n}+\sqrt{a_{n+1}}}$

Now, $\frac{1}{\sqrt{a_1}+\sqrt{a_2}} = \frac{\sqrt{a_2}-\sqrt{a_1}}{(\sqrt{a_2}+\sqrt{a_1})(\sqrt{a_2}-\sqrt{a_1})}$ (Multiplying numerator and denominator by $\sqrt{a_2} - \sqrt{a_1}$)

$$= \frac{\sqrt{a_2}-\sqrt{a_1}}{(a_2-a_1)}$$

$$= \frac{\sqrt{a_2}-\sqrt{a_1}}{d} \quad (\text{where } d \text{ is the common difference})$$

Similarly, $\frac{1}{\sqrt{a_2}+\sqrt{a_3}} = \frac{\sqrt{a_3}-\sqrt{a_2}}{d}$ and so on.

Then the expression $\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_n}+\sqrt{a_{n+1}}}$

can be written as $\frac{1}{d}(\sqrt{a_2} - \sqrt{a_1} + \sqrt{a_3} - \sqrt{a_2} + \dots + \sqrt{a_{n+1}} - \sqrt{a_n})$

$$= \frac{n}{nd}(\sqrt{a_{n+1}} - \sqrt{a_1}) \quad (\text{Multiplying both numerator and denominator by } n)$$

$$= \frac{n(\sqrt{a_{n+1}} - \sqrt{a_1})}{a_{n+1} - a_1} \quad (a_{n+1} - a_1 = nd)$$

$$= \frac{n}{\sqrt{a_1} + \sqrt{a_{n+1}}}$$

 VIDEO SOLUTION

69. D

Let the first term of the GP be "a". Now from the question we can show that

$$ar^{m-1} = \frac{3}{4} \quad ar^{n-1} = 12$$

Dividing both the equations we get $r^{m-1-n+1} = \frac{1}{16}$ or $r^{m-n} = 16^{-1}$ or $r^{n-m} = 16$

So for the minimum possible value we take Now give minimum possible value to "r" i.e -4 and n-m=2

Hence minimum possible value of $r+n-m=-4+2=-2$

 VIDEO SOLUTION

70. 548

General term = $38 + (n-1)17 = 17n + 21 = 17(n+1) + 4 = 17k + 4$

Each term is in the form of $17k + 4$

Least 3-digit number in the form of $17k + 4$ is at $k = 6$, i.e. 106

Highest 3-digit number in the form of $17k + 4$ is at $k = 58$, i.e. 990

106, 123, 140,, 990

$990 = 106 + 17(n-1)$

$n = 53$

Sum = $\frac{53}{2} (106 + 990) = 53 \times 548$

Average = $53 \times \frac{548}{53} = 548$

 VIDEO SOLUTION

Important Verbal Ability Questions for CAT (Download PDF)

71. A

$$\frac{1}{1-x} + \frac{1}{1+x} + \frac{2}{1+x^2} + \frac{4}{1+x^4}$$

$$\text{or } \frac{2}{1-x^2} + \frac{2}{1+x^2} + \frac{4}{1+x^4}$$

$$\text{or } \frac{4}{1-x^4} + \frac{4}{1+x^4}$$

$$\text{or } \frac{8}{1-x^8}$$

 VIDEO SOLUTION

72. 217

We are told that, for any natural number n_1 let a_n be the largest integer not exceeding $\sqrt{n}$

So for $n=1$, the largest integer not exceeding $\sqrt{1}$ will be 1

For $n=2$, the largest integer not exceeding $\sqrt{2}$ will be 1

For $n=3$, the largest integer not exceeding $\sqrt{3}$ will be 1

For $n=4$, the largest integer not exceeding $\sqrt{4}$ will be 2

We see a pattern here regarding the squares of the numbers,

Listing down all the perfect squares,

1, 4, 9, 16, 25, 36, 49, 64, ...

We see that the difference between 4 and 1 is 3 and there were three natural numbers in the given pattern with the value as 1,

So we can write for the rest of the numbers as well,

3 numbers will have value 1, giving a total value of 3

5 numbers will have value 2, giving a total value of 10

7 numbers will have value 3, giving a total value of 21

9 numbers will have value 4, giving a total value of 36

11 numbers will have value 5, giving a total value of 55

13 numbers will have value 6, giving a total value of 78

Now, only the values of a_{49} , a_{50} will have the value of 7, total value of 14.

Adding these values, we get the total sum as 217, which is the answer.

 VIDEO SOLUTION